CRANSEDS CHURRO

Preliminary Design Review

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1 Introduction

CranSEDS, Cranfield University's local chapter of UKSEDS, unites students passionate about advancing space exploration. The association fosters practical learning and innovation in the space sector by organizing competitions in rocketry, robotics, and satellite design.

The Propulsion Team builds on the success of the Hybrid Rocket Motor Test Stand (HRMTS) project "DRAKE," which earned first place in the Hybrid category at the 2024 edition of the prestigious Race2Space competition. This year, the team is focused on preparing for a full-stack test and developing a flight-ready hybrid rocket engine for EUROC 2025. The goal is to leverage the knowledge and capabilities established this year to transition towards the development of a liquid bi-propellant engine in future iterations.



Figure 1 - CHURRO Patch (Credit: Rachel Webb)

CHURRO (CranSEDS Hybrid Ultimate Rocket Research Operation) is dedicated to inspiring interest and enhancing student expertise in space propulsion, offering a platform for CranSEDS members to engage in hands-on rocket motor development. Moreover, it aims to support research initiatives and thesis projects for Cranfield and potentially other university students, utilizing the hybrid engine as a foundation. Launched in 2020, the

initiative envisions the integration of hybrid motors into flight systems and eventual progression to liquid propulsion technologies.

The work of the 2022-2024 teams has laid a robust groundwork for rocket engine development at Cranfield University. Their contributions in data collection and equipment acquisition have enabled this year's team to refine the development cycle and push the boundaries of performance for Race2Space and EUROC.

The CranSEDS High Altitude team will be designing their EUROC rocket so that it shares features with their Mach-25 rocket. They are currently focused on the Mach-25 project, which itself is only in the early stages of it's design. This rocket is shown below in Figure 2. With the submission of this PDR, both the High Altitude and Propulsion teams will have the beginnings of a design and intense collaboration will begin in the new year.

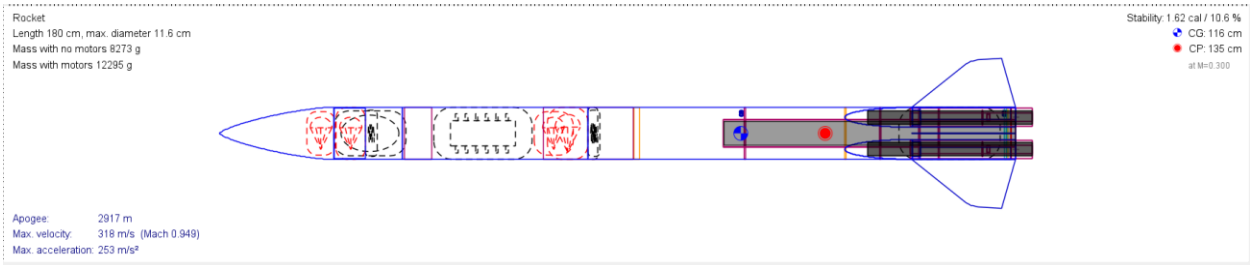


Figure 2 CRANSEDS MACH-25 ROCKET

2 Mission Statement

The CranSEDS Propulsion Team advances space propulsion by developing a hybrid rocket engine to propel the EUROC launcher to a 3 km apogee, fully meeting EUROC requirements.

The aim is to design the engine to be modular to enable seamless fitting into the launch vehicle being designed for EUROC. Meeting requirements such as combustion chamber pressure tests, failsafe design of loading lines, venting and static hot-fire tests will ensure the engine meets safety standards for the full-stack test in Race2Space as well as EUROC.

Building on past achievements, we aim to inspire innovation, support student research, and lay the groundwork for future liquid bi-propellant engine development at Cranfield University.

3 Requirements

To accomplish our primary mission, we systematically defined and allocated requirements across each subsystem, ensuring alignment with overall objectives.

3.1 Fuel and Combustion Chamber

Table 1: Fuel and Combustion Chamber Driving Requirements

ID	Requirements
LF-1	CHURRO shall achieve a total Impulse of 2,650.01 – 5,120 Ns
LF-2	The Combustion Chamber Pressure shall be less than 5.1 MPa
LF-3	The Combustion Chamber shall not exceed 1800 Degrees Kelvin
LF-4	The Oxidizer shall be Nitrous Oxide (N20)
LF-5	The injected oxidizer temperature shall be less than 36.42 degrees C
LF-6	The chamber shall not chemically corrode when in use
LF-7	The propellant should be inert when not in operation
LF-8	The fuel erosion rate must not exceed TBD micro meters per burn
LF-9	The Combustion Chamber outer diameter cannot be greater than 6 inches.

3.2 Valves and plumbing

Table 2: Valves and Plumbing Driving Requirements

ID	Requirements
VP-1	Operating Pressure: Up to 60 bar (adapted for Mojave Sphinx style).
VP-2	Temperature Range: Cryogenic to ambient (define exact range).
VP-3	Flow Rate: 5 kg/s oxidiser flow rate.
VP-4	Materials: Stainless steel or aluminium alloys, compatible with N_2O.
VP-5	Sealing: Use O-rings. soft-seal or metal-to-metal sealing, rated for cryogenic use.
VP-6	Actuation: Electrically or manually.

3.3 Electronics

3.3.1 Ignitor

Table 3: Ignitor Driving Requirements

ID	Requirements	Target Value
RQ-IGN-001	Input voltage range for power supply.	12-24V DC
RQ-IGN-002	Input current during ignition.	Peak 5A (TBD)
RQ-IGN-003	Output voltage of the ignitor circuit.	15kV ± 5%
RQ-IGN-004	Maximum current output to the arc.	10mA (TBD)
RQ-IGN-005	Insulation rating for high-voltage lines and components.	20kV (TBC)
RQ-IGN-006	Response time from trigger signal to ignition arc.	≤100ms (TBD)

RQ-IGN-007	Isolation gap between high-voltage and low-voltage sections.	≥10mm
RQ-IGN-008	Operational temperature range.	-20°C to 50°C (TBC)
RQ-IGN-009	Vibration tolerance.	10g RMS (TBD)
RQ-IGN-010	Shock tolerance.	15g (TBD)
RQ-IGN-011	Energy storage capacitance rating.	15kV, 10nF (TBC)
RQ-IGN-012	Compactness of the ignitor circuit.	≤10x10x5 cm
RQ-IGN-013	Weight of the ignitor circuit.	≤300g (TBC)
RQ-IGN-014	Moisture protection for outdoor or humid conditions.	IP67 or conformal coating (TBD)
RQ-IGN-015	Voltage regulation stability at 15kV.	±5% (TBC)
RQ-IGN-016	Indicators for armed, active, and fault states.	LED Indicators
RQ-IGN-017	Digital communication signal level for trigger.	5V TTL
RQ-IGN-018	Safety cutoff current.	10A (TBC)
RQ-IGN-019	Pre-launch testing functionality.	High-voltage output test mode (TBD)
RQ-IGN-020	Arc chamber material for containment.	Ceramic or high-strength glass (TBC)

3.3.2 Control systems

Table 4: Control Systems Driving Requirements

ID	Statement	Rationale	Type
FUN-CS-002	The control system must control the build-up of thrust at increments of TBD N/ms from 0% to TBD% Thrust during start-up sequence.	To prevent high temperature spikes inside the combustion chamber	Functional
FUN-CS-003	The control system shall keep the desired chamber pressure of TBD during the operational sequence.	To achieved desired engine performance	Functional
FUN-CS-006	The control system shall provide fuel-rich cutoff in the combustion chamber during cut-off sequence.	This prevents damaging temperature spikes and results in smooth and rapid thrust-termination.	Functional

PER-CS-001	The chamber pressure shall have a rise time of TBD seconds (±TBD ms).	Controlling the rise time will prevent pressure and temperature spikes	Performance
PER-CS-002	The chamber pressure shall not overshoot over TBD bar (±TBD Pa).	It's important to control the overshoot where it doesn't exceed the stress limits of the combustion chamber	Performance
PER-CS-004	The chamber pressure shall reach a integral square error of TBD (±TBD)	This will define how accurate and stable the control is.	Performance
SAF-CS-001	The control system must provide a capability of system shutdown if the chamber pressure exceeds TBD bar.	To prevent unplanned rapid disassembly a limit of chamber pressure needs to be defined	Safety
ECSS_CS_004	The parameters that allow monitoring and controlling the propulsion system shall be defined including their corridors and accuracy. NOTE 1 Measurements which are necessary to meet safety requirements are of particular importance. NOTE 2 When used, the functions of the monitoring and control system can include: Monitoring the state of a subsystem or system. Collecting information for further processing, e.g. transmission to ground. Comparing the state of the subsystem or system with the intended one. Activating equipment to suppress deviations from the intended state of the subsystem or system.	N/A	ECSS

3.4 Mechanics

Table 5: Mechanics Driving Requirements

ID	Requirements	Rationale
M-TL-1	The combustion chamber design shall be validated for thermal performance and material compatibility.	Ensures efficient combustion and durability under high temperatures, guided by ECSS-E-ST-32 and material standards.
M-TL-2	The nozzle design shall be validated for high-temperature and mechanical stresses during operation.	Ensures nozzle durability and performance during thrust generation, following ECSS and aerospace standards for nozzle design.
M-TL-3	The tank material shall be compatible with the oxidizer.	Ensures long-term reliability and safety; based on material compatibility standards (e.g., ASTM G93, ECSS-Q-ST-70).
M-TL-4	The combustion chamber and nozzle shall have proper heat insulation, cooling design, or material selection.	Prevents overheating and structural failure, ensuring sustained performance; guided by ECSS and industry best practices.
M-TL-5	All components shall include secure integration points for valves and piping.	Allows for robust assembly and operational reliability, following standard interfacing methods in aerospace (e.g., ASME, ECSS).

M-TL-6	All components shall withstand maximum operating pressure with an appropriate safety factor.	Ensures structural integrity during flight, referencing aerospace safety standards (e.g., NASA-STD-5012, ECSS).
M-TL-7	All designs shall be validated for mechanical stresses (hoop, axial, thermal) by theoretical calculation and FEA.	Ensures that the design can withstand operational stresses; based on ECSS and industry standards for chamber design validation.
M-TL-8	The piping system shall be designed to prevent leaks and withstand pressurized fluid flow.	Ensures safety and reliability in fluid transfer, following ECSS and industry standards for piping integrity.
M-TL-9	The entire system shall meet General Safety Standards, including pressure relief and emergency shutdown.	Ensures safety during operation, following ECSS and safety standards (e.g., ECSS-Q-ST-40).
M-TL-10	The on-board components shall not exceed the specified mass limitation.	Ensures compliance with the rocket's mass budget, optimizing performance and structural safety. Refer to competition-specific payload and mass distribution rules.

4 Timeline and Team Structure

The 2024/25 CranSEDS Propulsion team consists of 31 members. These students come mainly from the Thermal Power and Propulsion and Astronautics and Space Engineering courses at Cranfield. The team is led by two project manager and is subdivided into several work packages as shown in Figure 2 below. Each of these work packages has been allocated a Team Leader who responsible for each team's progress and provided the PM's a point of contact within each team. Members are free to work on as many teams as their university schedule allows, however everyone is assigned one primary team based on their own interests. Especially important is the Integration team which is tasked with liaising with the CranSEDS High Altitude team.

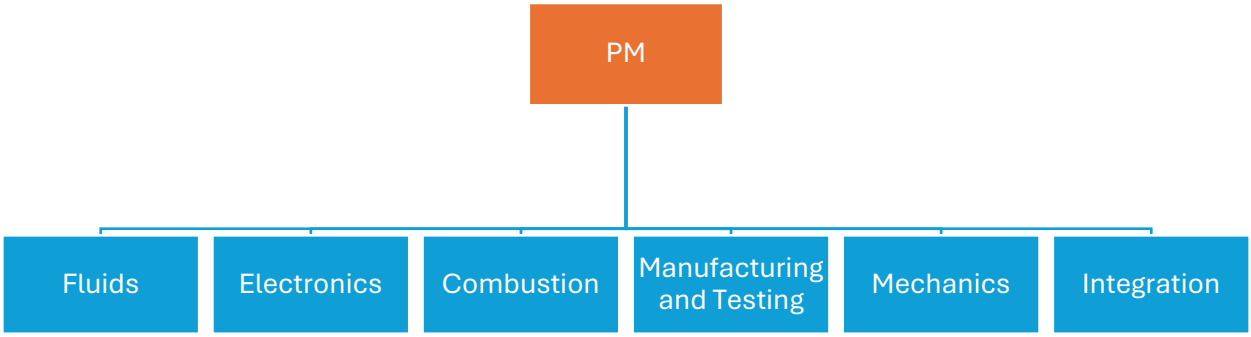
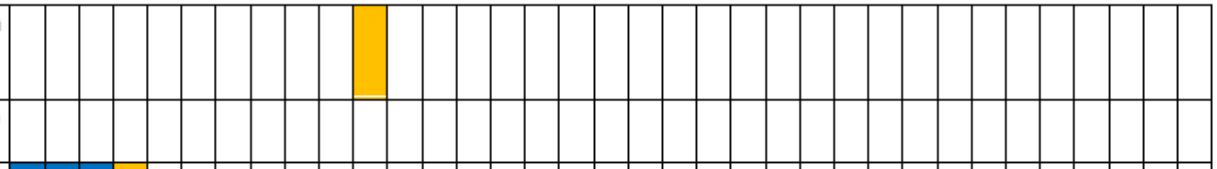
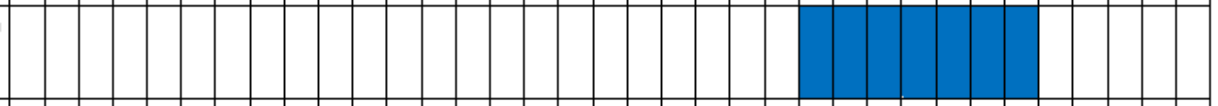
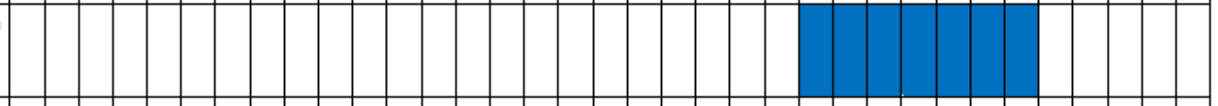
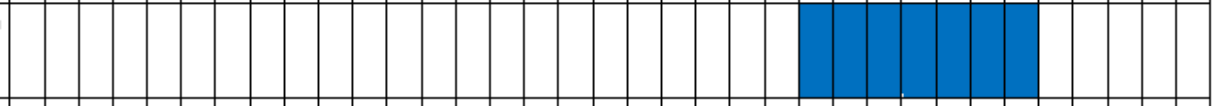
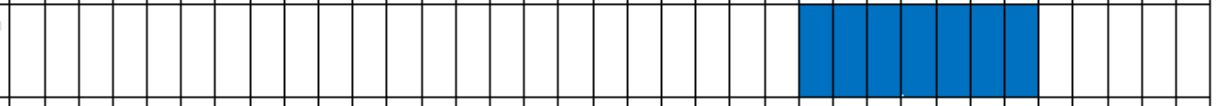


Figure 3: Team structure

Project:	CranSEDS Hybrid Rocket 'CHURRO'																																							
Deadlines																																									
					December				January					February				March					April					May					June					July	
Milestone name	Assigned to	Start	Finish		W2	W3	W4	W5	W1	W2	W3	W4	W5	W1	W2	W3	W4	W1	W2	W3	W4	W5	W1	W2	W3	W4	W5	W1	W2	W3	W4	W5	W1	W2	W3	W4	W5	W1	W2	W3	W4	W5	W1	W2
General																																									
Initial Requirements	PMs	15/12/24	20/12/24	Planed																																						
		15/12/24	20/12/24	Done																																						
Systems Requirements	PMs and systems leaders	20/12/24	31/12/24	Planed																																						
		20/12/24	31/12/24	Done																																						
Handover meeting	PMs and mentor	01/12/24	01/12/24	Planed																																						
		01/12/24	01/12/24	Done																																						
System Research	Systems Leaders	20/12/24	31/12/24	Planed																																						
		20/12/24	31/12/24	Done																																						
Concept development	All	20/12/24	01/02/25	Planed																																						
		20/12/24		Done																																						

Race2space workshop	ALL		14/01/25	Planed	
				Done	
PDR	PMs		12/31/24	Planed	
			12/31/24	Done	
CDR			25/04/25	Planed	
				Done	
R2S	All	23/06/25	23/06/25	Planed	
				Done	
Manufacturing					
Material Selection	Mechanics Lead	05/01/25	01/02/25	Planed	
				Done	
CAD	Mechanics Lead	10/01/25	10/02/25	Planed	
				Done	
BOOM (Blueprints)	Systems Leaders	10/02/25	15/02/25	Planed	
				Done	

Contact suppliers	PMs	01/02/25	02/04/25	Planed	
				Done	
Manufacturing	All	20/02/25	01/05/25	Planed	
				Done	
Testing					
Pressure testing	PMs and testing	01/05/25	tbd	Planed	
				Done	
Injector flow testing	PMs and testing	10/05/25	tbd	Planed	
				Done	
Hot/cold testing	PMs and testing	10/05/25	tbd	Planed	
				Done	
Electronics	PMs and testing	10/05/25	tbd	Planed	
				Done	

Part	No	Item Cost(£)	
Tank	1	400	400
Fuel	1	100	100
Oxidiser		0	0 (R2S)
Nozzle	3	200	600
Igniter	2	75	150
Valves	2	100	200
DAQ	1	500	500
CC	1	110	110
Machinin	1	800	800
			0
Total			2860

The preliminary budget hasn´t has not exceeded the limit of CranSEDS intended budget, so it will not be necessary to raise more money

5 Detailed Design

5.1 Overview of Design Process

For the purpose of this PDR each team has been allowed somewhat independently research their individual area. This approach has been followed so that the team members will have a solid foundation of knowledge of every component of the rocket. One of the issues we face at CranSEDS is the lack institutional knowledge as many members are only on campus for 1 year only. However, by allowing for such a wide design space is the conclusions reached by each team may not be compatible. Errors have also arisen due to false assumptions as the team gains knowledge of the system.

However, this work will form a solid foundation onto which the hybrid motor will be built. In early 2025, the designs presented below will be iterated upon so that they may fit into an overall system.

5.2 Preliminary Sizing

To gain an initial estimate of the size of the rocket two methods were used to size the rocket. The following formulas were taken from Newlands (2017) and were used first:

$$\Delta e_{potential} = \frac{\mu}{Re + A}$$

$$Mass\ Fraction = \exp\left(\frac{\Delta V}{C_e}\right)$$

$$Mass\ Ratio - 1 = \frac{Propellant\ Mass}{Empty\ Mass}$$

The following formulas from Humble(2007) were also used:

$$m_{inert} = \frac{f_{inert}}{1 - f_{inert}} m_{prop}$$

$$m_{fuel} = \frac{m_{prop}}{1 + O/F}$$

$$m_{ox} = m_{prop} - m_{fuel}$$

$$\dot{m}_{prop} = \frac{F}{I_s g_o}$$

$$\dot{m}_{fuel} = \frac{\dot{m}_{prop}}{1 + O/F}$$

$$\dot{m}_{ox} = \dot{m}_{prop} - \dot{m}_{fuel}$$

The values used in these formulas are given in the Table 6 below:

Table 6

Property	Value	Justification
μ	39860500	As given in Newlands(2017)
Earth Radius (Re) (km)	6378.14	As given in Newlands(2017)
Sea Level Gravity g_o (m/s^2)	9.81	
Empty Mass (kg)	10	Based on early work of EUROC Team
Apogee (M)	3500	Conservative estimate
Inert Mass Fraction (Finert)	0.2	As given in Humble(2007)
Thrust (F) (N)	2000	Based on early work of EUROC Team
Isp (s)	204.26	From CranSEDS 2024 report
Exhaust Velocity (Ce) (m/s)	2682	As given in Newlands(2017)
O/F Ratio	7:1	As given in Newlands(2017)

These formulas were then written into a python script which can be found in Appendix D.

The results are shown below in Table 7:

Table 7

Result	Newlands	Humble
Delta V (m/s)	450.3639	-
Fuel Mass(kg)	0.7934	1.4344
Oxidiser Mass(kg)	1.034	1.2551
Propellant Mass (kg)	1.8284	2.6896
Fuel Mass Flow (kg/s)	-	0.4657
Oxidiser Mass Flow(kg/s)	-	1.2551
Prop Mass Flow (kg/s)	-	0.9981

The discrepancy is likely due to the fact that a given inert mass fraction of 0.2 was present in the formulas used in Humble(2007). The formulas from Newland(2017) do not include this.

In conclusion, these calculations were undertaken to gain a rough understanding of the overall dimensions and propellant masses of the rocket that is being designed. Different sets of formulas were used as Humble(2007) provided a slightly more accurate result.

5.3 Fuel Selection

The primary objective of CHURRO is to integrate the engine into the CranSEDs EuROC 2025 launch vehicle. As a result, It has been set that CHURRO must achieve a total

Impulse (I) equivalent to 5,120.01–10,240 Ns, M Class to reach the 3km altitude of the EuROC competition. Iterating on last year's design, CHURRO follows a liquid oxidizer and solid fuel hybrid system.

$$C^* = \frac{1}{\gamma}\left(\frac{\gamma+1}{2}\right)^{\frac{\gamma+1}{2(\gamma-1)}}\sqrt{\frac{\mathcal{R}T_c}{mol}}$$

$$I = (I_{sp})m = \ (C^*\ C_f)m$$

For the fuel selection team to achieve this, the use of C* and Cf approach was used to isolate the fuel selection from the engine geometry (Humble, 2007). Through the equation above a specific impulse can be defined. This, as well as last year's 16 kg total mass, can be used to calculate the total impulse. This will define if the design is valid to meet the requirements. As the following section concerns fuel selection, theoretical, which assumes chemical equilibrium conditions, will be used to define the performance of candidate fuels. Other considered factors include:

- Toxicity
- Cost of Fuel Materials
- Cost of Manufacturing (if required)
- Handling and Safety
- Chamber Combustion Temperature
- Available Data/ Heritage

The goal of CHURRO is to iterate on last year's design and integrate it into a launch vehicle, as a result, Nitrous Oxide was selected as the oxidizer. This is due to the self-pressurizing properties of the gas at room temperature (Newlands). This simplifies tank design and reduces the structure mass required of CHURRO (Newlands).

5.3.1 Requirements discussion

Table 1 defines the driving requirements. L1-F1 was derived from the requirements place on CHURRO by the EuROC team. L1-F2 places an upper limit on the chamber pressure. The maximum chamber pressure must be significantly lower than 5.1 MPa as that the vapor pressure of Nitrous Oxide. For sufficient mixing, the Nitrous oxide should inject as a liquid as it allows for complete atomization of the oxidizer. L1-F3 is a critical requirement, as seen in later sections, this requirement defines the maximum allowable chamber temperature for different combustion chamber materials and fuels. Therefore, it places an upper limit of based on maximum chamber temperature as the fuel.

Therefore, LF-3 summarizes a key constraint into the fuel selection choice: the combustion chamber temperature must be feasible, the combustion chamber doesn't melt.

LF-3 also defines a key design trade-off: the of fuel mix must be high enough to achieve LF-1, but it mustn't be too extreme to cause the combustion chamber wall to fail. This trade-off informs the O/F choice. A fixed material approach was taken as the aim of the engine is to achieve high thrust-weight ratio's, therefore inert mass should be minimized.

5.3.2 Initial Fuel Identification

Initial Fuel Identification was carried out through a literature review, followed by a pugh decision matrix. It was decided that CHURRO's ideal fuel would hold these characteristics:

- Easy to Manufacture
- Storable and non-toxic
- Low cost per kilogram
- Non-harmful combustion products
- Good Insulator
- High characteristic Velocity ()

As a student design team, our biggest constraints are the lack of safety expertise and cost of the Hybrid engine. Moreover, due to the nature of the project, heat transfer properties were important to ensure that an uncontrolled explosion wouldn't occur, and the thrust to weight ratio be sufficient to reach 3km. As a result, fuel density, material and manufacture cost, and storability were prioritized over propellant characteristic velocity and ignition reliability.

This is shown in Table 7, cost of material and safety/handling have a combined weight of 0.60 out of 1, while the ignition and burn performance had a combined weight of 0.15 out of 1. The consequence of this is show in the table below, High Density Polyelene (HDPE), is shown to have a higher utility score than Hydroxyl Terminated poly-butatiene (HTPB), despite HTPB being the Hybrid industry standard being defined in Humble as the comparable to RP-1 for liquid rocket engines.

Paraffin wax has the highest utility score of all candidate fuels of 8.2. This is due to several factors: high regression rate of the fuel allows for simpler single port grain design reducing manufacturing complexity and reducing manufacturing cost. Furthermore, Its inert and

cost-effective. This presents paraffin wax as a good fuel was to prioritize performance and minimize storability hazards. In other words, its ideal for static thrust tests by student teams. However, since CHURRO will be integrated into a launch vehicle, it must prioritize heat transfer and reliability rather than performance. This is paraffins wax weakness; paraffin cracks easily under mechanical stress which cause large temperature spikes across the combustion chamber wall, and it could fail (Riccardo Bisin, 2023). As a result, paraffin's performance potential and storability carry it forward to O/F analysis however it is not the first choice of fuel now.

HTPB is a high performance, industry standard fuel, comparable to RP-1 on performance. It's storable, easy to manufacture and holds significant hybrid heritage. However, it is 10x more expensive than paraffin wax or HDPE. In an event that HDPE or Paraffin wax do not provide the necessary specific impulse, HTPB will be used to due to the high performance potential.

HDPE offers similar performance to paraffin wax without the risk of cracking and ununiform heat transfer to the combustion chamber wall. This presents it as an ideal fuel to be taken forward into quantitative analysis.

5.3.3 O/F selection

O/F selection used NASA's CEA to determine different at different oxidizer to fuel ratios. CEA code posits chemical reactions across the shock wave and then minimizes the Gibbs free energy to determine thermodynamic properties at chemical equilibrium. As shown in the graph below, high O/F leads to high, therefore high till O/F of 6.

Table 8: Available reactants

Reactant	**T0 (K)**	**Hf (kJ/mol)**
N20	298	81.6
HDPE	288	-28.18
Paraffin Wax	288	-501.00

The conservatism of the steady state heat transfer equations dismisses the need of a maximum chamber temperature margin. This presents an inherent margin of the subsequent analysis as the burn time of a hybrid engine is around 5-10 seconds. This is too short for steady state conditions to be reached.

It was assumed the grain geometry was a simple single port and cylindrical grain geometry, the grain would be burned in the direction from the center of the combustion chamber to the wall. As a result, the O/F will decrease over time.

From the figures, O/F of interest are 2.1 and 3 for both HDPE and Paraffin. Under steady state heat transfer equations, an O/F of 2.1 is within the maximum O/F allowed before walls of combustions chamber start to fail. A high O/F is desirable as it enables higher mass-flow and therefore thrust. This is crucial to maximize thrust to weight ratio. It also enables a high C*, therefore Isp.

An O/F of 3 was also chosen as a possible operating O/F, this is the optimal C* while aiming to reduce chamber temperature. The figure shows It lays outside the allowable range of O/Fs. However, as transient conditions are implemented it is expected the maximum chamber temperature will increase and the optimal O/F of 3 may be utilized.

Due to the suspected grain geometry, the O/F mixture is suspected to decrease as the burn elapses. This means that the initial O/F selection determines the maximum chamber temperature.

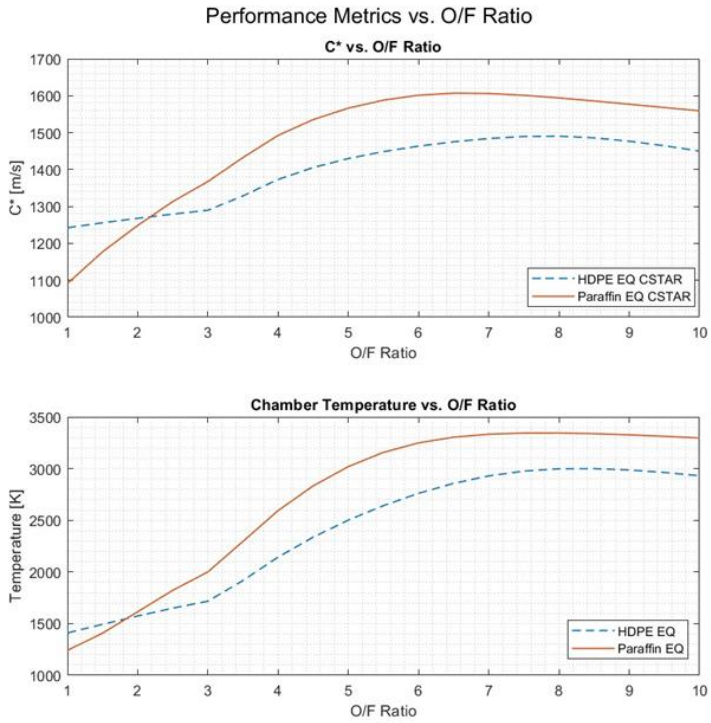


Figure 4: Mixture ratio effect on Combustion

Table 9: Possible operating points

O/F (-)	Characteristic Velocity (ms-1)	Combustion Temperature (K)

HDPE: 2.1	1268.0	1574.18
Paraffin: 2.1	1248.6	1614.89
HDPE: 3	1289.9	1717.38
Paraffin: 3	1367.6	2000.63
HDPE: 6.5	1475.2	2856.41
Paraffin: 6.5	1607.0	3304.25

5.3.4 Future Work

Although this serves as an appropriate first approximation, to identify a single fuel at a single mixture ratio, a transient heat transfer model will be defined. This will reduce the margin inherent in the heat transfer calculations present in Figure 3 7. A refined heat transfer model will also quantify the effect of paraffin cracking in the combustion chamber wall temperature, therefore the risk it possesses as a fuel choice.

Insulation liner thickness has significant consequences on the chamber wall temperature, therefore the O/F selection. As a result, a sensitivity analysis will be carried out which will identify the effect of the reduction of combustion chamber radius, and subsequent insulation thickness increase, on combustion performance.

5.4 Tank Design and Size

Assumptions made to calculate the design and size of the tank:

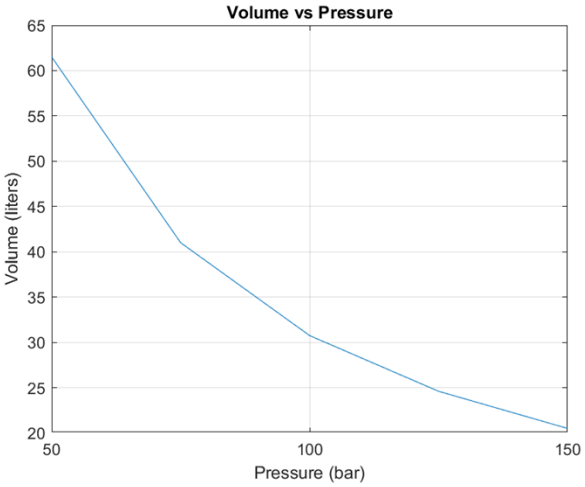
- Mass flow rate = 0.5 kg/s
- Chamber pressure should be around 3.5 MPa, there should be some losses through the fuelling system (the latest Cranfield report, calculated around 1 MPa in pressure losses) so I have set 5 MPa as the initial pressure to iterate.
- The temperature inside the tank is 300K
- We will be feeding the system for 10 seconds.
- The oxidizer considered is N2O

These assumptions have been obtained by looking at past reports

With these assumptions, the overall volume is calculated with the following equation:

$$V = \frac{\dot{m} \cdot t \cdot R \cdot T}{P M_{ox}}$$

With this equation, we can obtain the next figure, which shows the possible oxidisers tank. The blue line splits the possible tank volumes in 2 zones. If we chose a point in the superior region the tank will withstand our requirements.



From looking on the internet, here it is one tank that could be chosen

Capacity (L)	Diameter (mm)	Height (mm)	Pressure (MPa)	Weight (kg)
40	219	1325	15	48.5

5.5 Valves and Plumbing

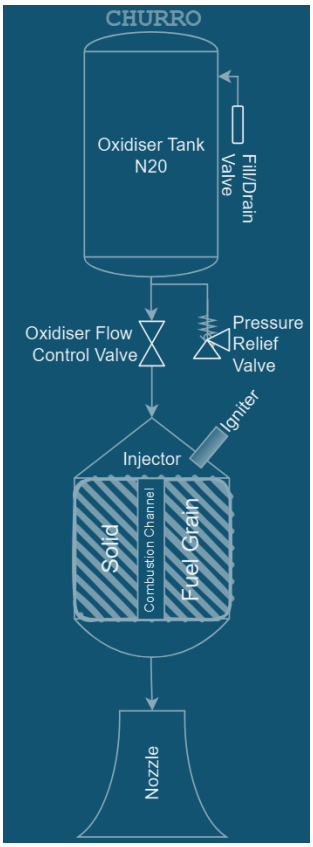


Figure 5 Feed system block diagram

The feed system, design must meet both top-level and technical requirements to ensure optimal performance, safety, and integration within the rocket's overall architecture. Feed system block diagram of CHURRO is shown in .

5.5.1 Requirements discussion

Valves and Their Functions

The system requires the following valves:

- Fill/Drain Valve: A ball or globe valve used to fill and drain the oxidiser tank. It can be manually operated or electrically actuated.
- Oxidiser Flow Control Valve: A solenoid or proportional valve to regulate oxidiser flow to the combustion chamber, offering precise control via electrical actuation.
- Pressure Relief Valve: A spring-loaded or pilot-operated valve to prevent over-pressurisation, designed to handle the maximum operating pressure.
- Check Valve: A one-way valve to prevent backflow into the oxidiser tank, enhancing system safety.

Pressure Relief Mechanisms

The recommended pressure relief mechanism combines a reusable pressure relief valve, a single-use rupture disc, and a static vent. This dual-layer safety system provides robust protection against over-pressurisation. While rupture discs are lightweight and fail-safe, they require replacement after activation. Pressure relief valves are reusable but more complex and require periodic maintenance.

Pressure Drop Estimates

The oxidiser flow control valve is estimated to experience a pressure drop of 1–2 bar, depending on its flow coefficient (Cv value). Pressure drops for the pressure relief valve and fill/drain valve are negligible under normal operations, with significant drops only occurring during filling or draining.

Testing and Verification Plan

A comprehensive testing plan ensures the system meets all performance, safety, and operational requirements:

- Functional Tests: Validate valve operation by testing open/close cycles and response times.
- Flow Tests: Measure pressure drops and flow rates to verify against design specifications.
- Pressure Relief Tests: Confirm the activation of relief mechanisms at the correct pressures.
- Cryogenic Tests: Test the system under sub-zero conditions to ensure material compatibility.
- Leak Tests: Perform leak testing under maximum pressure to ensure system integrity.

Risk Analysis

Risk	Impact	Mitigation
Valve Failure	Flow disruption or uncontrolled oxidiser flow.	Include redundancy or manual override options.
Over-Pressurisation	Catastrophic failure of tank/system.	Use dual-layer safety: relief valve + rupture disc.
Cryogenic Material Failure	Loss of integrity in extreme temperatures.	Use cryogenically rated materials.
Electrical Integration Issues	Delays in system operation or failure to actuate valves.	Early collaboration with electrical team.
Oxidiser Leaks	Safety hazard and performance degradation.	Conduct rigorous leak testing during assembly.

5.6 Nozzle Geometry

To design an adequate nozzle, it is essential to know the pressure and temperature conditions in the combustion chamber, as these determine the behaviour of the exhaust

gases and the system's efficiency. In this case, the following conditions have been assumed in the combustion chamber:

- Average pressure (Pc): **1.50 MPa**
- Average temperature (Tc): **2250 K**
- Average molecular mass of the combustion gases: **0.03 kg/mol**

These values were determined based on the selected fuels and oxidizers. With this data, the throat area can be calculated by using the following expression:

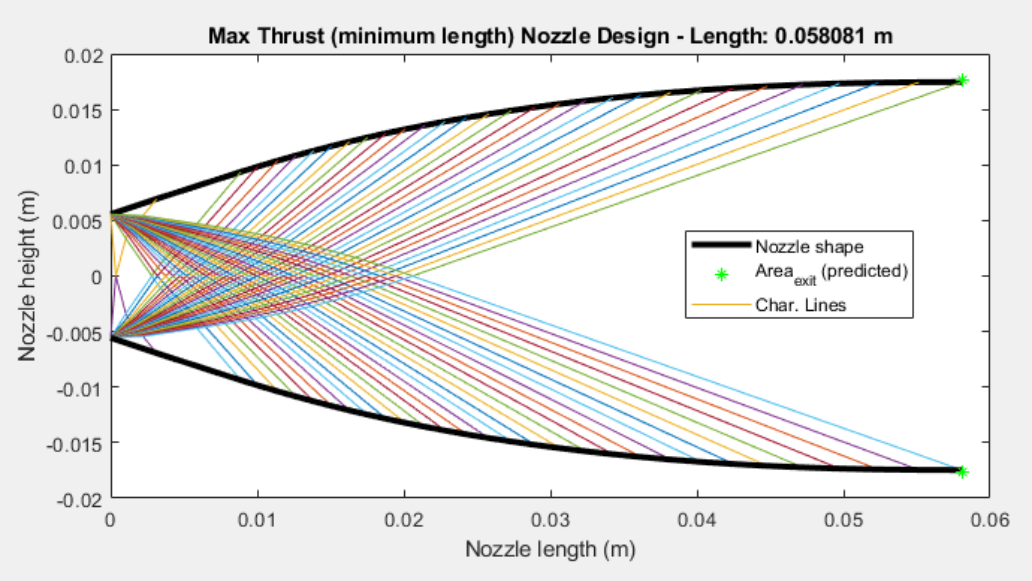
$$\frac{A}{A_*} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \frac{1}{M} \cdot \left[1 + \frac{\gamma - 1}{2} M^2 \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Knowing the required throat area, the geometry of the divergent section of the nozzle can be determined by making use of a MATLAB code. This code begins with essential input parameters, including the throat diameter, combustion chamber pressure, temperature, and gas properties such as the specific heat ratio and molecular weight. These inputs define the initial conditions for the nozzle design. The throat area is calculated based on the given diameter, representing the critical region where the flow is choked. Combustion chamber pressure and temperature provide the upstream conditions necessary for determining the expansion behaviour of the gases through the nozzle.

Using these inputs, the code designs the geometry of a nozzle’s divergent section with the Method of Characteristics (MOC). The throat serves as the critical reference point where the flow transitions from subsonic to choked conditions. From this starting point, the code calculates the expansion path of the gases by generating characteristic lines, which guide the shaping of the nozzle walls to ensure smooth and efficient supersonic expansion.

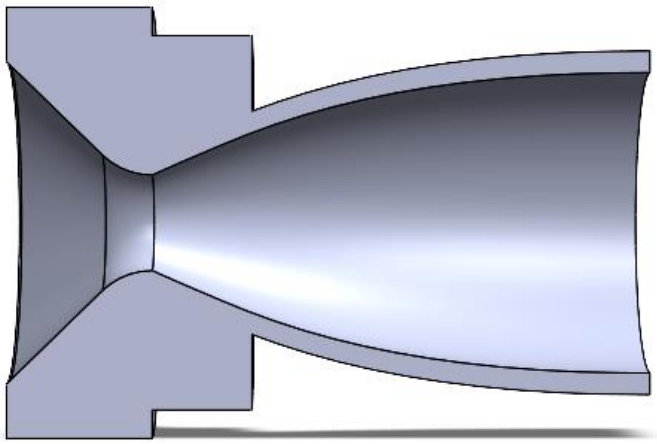
The procedure involves solving the governing equations for gas flow iteratively, calculating the Mach number and flow angles along a grid of characteristic lines. These lines are extended to form the nozzle contour, which is tailored to achieve the desired exit Mach number. The design optimizes the nozzle geometry to maximize thrust while minimizing its overall length, ensuring efficient gas expansion with minimal material use.

The process results in a fully defined nozzle profile, with precise calculations of the divergent section’s length, height, and exit area. For the conditions we have established, the following geometry and characteristics of the nozzle's divergent section are obtained:



Property	Value
Throat Diameter (m)	0.0111
Throat Area (m^2)	9.7037×10^{-5}
Exit Diameter (m)	0.0349
Exit Area (m^2)	9.5903×10^{-4}
Final Nozzle Length (m)	0.0581
Maximum Thrust (N)	227.5236
Thrust Coefficient (C_f)	1.5631

The MATLAB code includes a feature that exports the calculated nozzle geometry, specifically the coordinates of the divergent section, in a format suitable for CAD software. This allows for seamless integration of the computational results into modelling and simulation environments. In this case, the coordinates of the divergent section were saved in a CSV file, which can be imported into CAD tools such as SolidWorks:



The nozzle thickness and casing have been designed in a very preliminary manner, with arbitrary values assigned for now. Similarly, the convergent section of the nozzle has been designed in an equally preliminary fashion, following a simple approach.

The convergent section has been designed according to the bell-nozzle figure below, where convergent section begins with a radius of curvature of 1.5 times the throat radius to ensure smooth flow transitions. Then, the section was extended linearly from this point to connect with the combustion chamber.

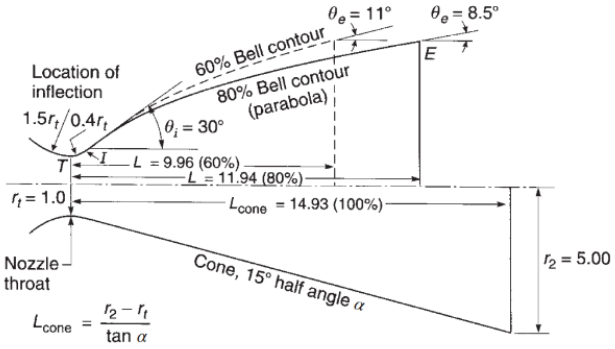


Image courtesy of Sutton GP and Biblarz O. Rocket Propulsion Elements, 9e. p77.

Generally, the nozzle configuration using basic geometry is developed using curves plotted empirically for nozzle efficiency vs the fraction of a 15° conical nozzle and initial/final parabola angles vs expansion ratio. However as elucidated above, the coordinates of the parabola are obtained from the MATLAB code and the nozzle profile is thus obtained.

Further work involves designing the thickness and casing of the nozzle with greater precision, considering the mechanical and thermal loads. Once chamber conditions are established, a decision can be made on the selection of the appropriate material/s for the converging section of the thrust chamber and the rest of the nozzle.

Additionally, the matings including sealing between the nozzle and the combustion chamber will be studied in greater detail. This involves refining the geometry and length of the convergent section to ensure it meets all combustion chamber requirements

5.7 Combustion Chamber

The combustion chamber work package aim is to design a combustion chamber that:

- Achieve a total Impulse of greater than 2,560.1 Ns
- Ignite reliably
- Provide enough acceleration to reach 30ms-1 at the end of the launch rail
- Ensure the combustion chamber temperature is acceptable for the desired chamber material.

Due to the transient nature of Hybrid engines, this can be broken down into four areas

- Fuel Grain and Port geometry, which studies the thrust profile of different grain configurations
- Ignition Methods, which studies different ignition operation methods.
- Heat Transfer characteristics, which conducts thermal analysis to ensure the combustion temperature will not melt the chamber wall.
- Fuel Selection, which studies combination of fuel combinations and mixing efficiency on C*.

5.7.1 Fuel Grain and Port Geometry

The aim of CHURRO is to be integrated into a launch vehicle, therefore the thrust profile must be like Figure 6, where there is a large thrust at the beginning of the thrust curve. This thrust should provide enough acceleration so, at the end of the EuROC launch rail, the launch vehicle will achieve a velocity greater than 30ms-1. Beyond that, there are no constraints on the grain geometry however a simple configuration is preferred for ease of manufacturing. As seen in Figure 2, this means all thrust profiles are applicable except progressive tubular designs.

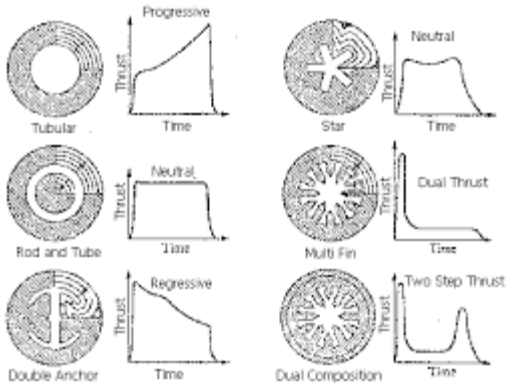


Figure 6: Thrust profiles (Ceyhun Tola, 2016)

There is more effective thrust at higher altitudes because the atmospheric pressure decreases with altitude due to equation below. Therefore, to maximize performance, a two-step thrust profile to favourable to guarantee high launch acceleration and high altitude required by EuROC. However, the intricacy of the grain cross section places a demand on casting. For paraffin wax, this could present a challenge as paraffin wax needs to be cast so that there are no voids (al).

$$F = \dot{m}u_e + (P_e - P_a)A_e$$

Neutral and Regressive thrust profiles present easier manufacturing processes. This makes it suitable for a paraffin wax fuel where the casing process is important (al). However as shown in equation above, the effective thrust my lower over the burn time due to thrust according at approximately sea level. Neutral profiles place a higher importance on the performance of the engine as the launch thrust is less than a regressive profile.

Therefore, a future work involves a trade-off study to be conducted between the thrust profiles and the available fuel choices described earlier. This will require quantification of regression rates of the fuels, and an estimate on erosive burning. Grain shape will also be analysed with respect to its insulation properties. Further Testing may be required to define regression coefficients accurately.

As CHURRO is an iteration of last years, design a single port hole is acceptable at his stage but is liable to change as the trade-off study is conducted.

5.7.2 Combustion Chamber Parameters

The outer radius of the combustion chamber was first approximated to the inner radius of the EuROC launch vehicle, 6 inches. The inner radius of the combustion chamber is the summation of the wall and linear thickness subtracted from the outer radius. The inner radius is 127mm however is subject to change as the grain geometry and liner thickness choices are clearer.

The combustion chamber length is summation of the characteristic length, grain length required, and the aft mixing chamber length. This is yet to be determined as grain shape has not been decided. Once grain geometry is sufficient to achieve the thrust required, the chamber length can be calculated.

The Chamber pressure was chosen to be 1.5 MPa. The previous years hybrid engine chamber pressure was 3.1 MPa. The change in operation chamber pressure is due to the material change from last year as the team is unsure how the temperature affects the young's modulus of the combustion chamber wall. As safety is the priority a conservative approach was taken.

5.7.3 Heat Transfer Characteristics

The aim of the Heat transfer work package is to determine whether the combustion chamber temperature is acceptable so that the combustion wall does not melt. As one of CHURRO's figure of merit is thrust to weight ratio, this is especially important as the lightweight materials are used. These materials have low melting temperatures.

The preliminary Heat Transfer model assumed a steady state, adiabatic, heat transfer scenario with convection and conduction throughout the chamber. As the fuel grain geometry has not been decided it was assumed a rod and tube design was sufficient. Other assumptions include:

- Combustion Chamber centre, T1, is the combustion temperature
- The fuel, insulation liner and metal casing all conduct heat uniformly
- The Combustion chamber material is an copper alloy and melts at approximately 993.5 Kelvin.

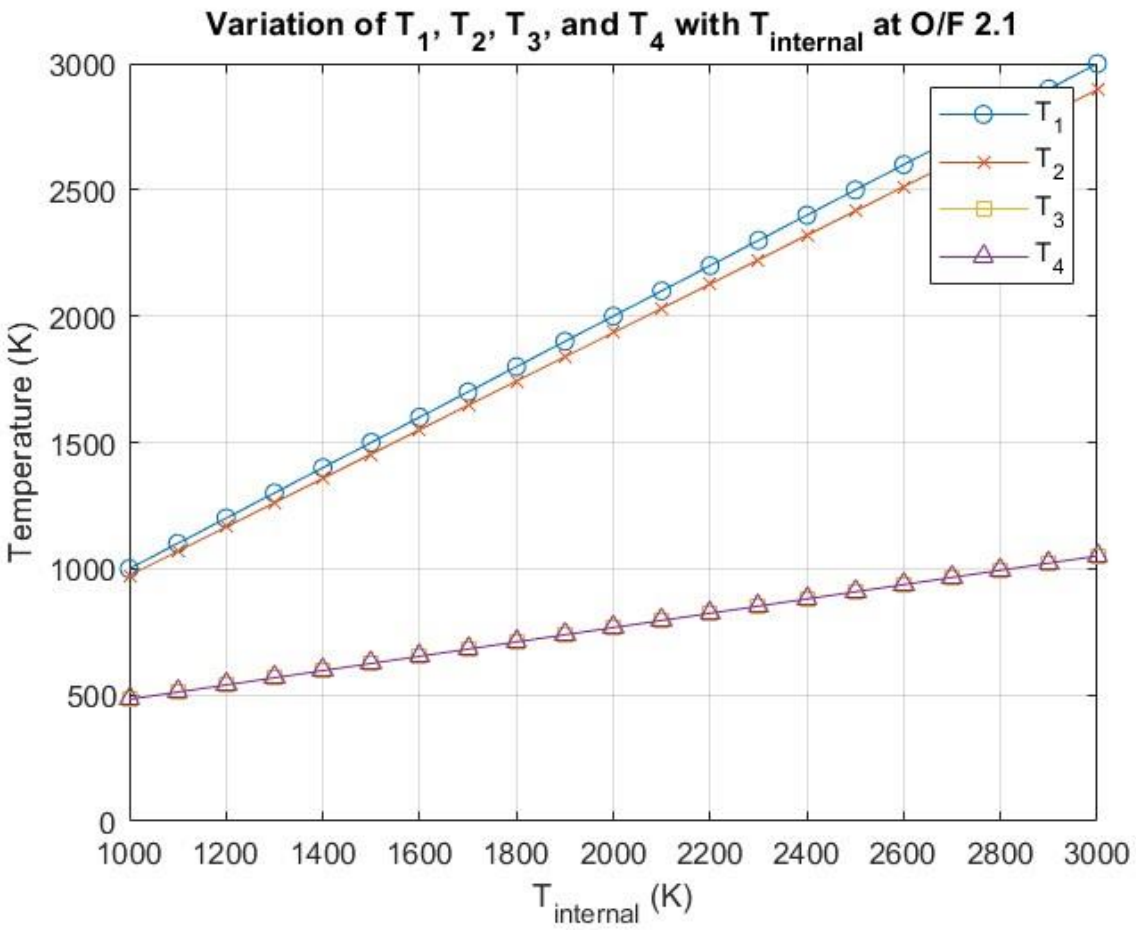


Figure 3 7: Heat Transfer from centre to chamber wall

Figure 3 shows how the internal temperature of the combustion affects the wall temperature, T4. It can be observed that a chamber temperature of < 2900 K is below the melting temperature of the copper alloy of 993.5 Kelvin. However, young's modulus reduces with wall temperature. This is important as it could explode if it reduces too much. Therefore, a wall temperature of 700 Kelvin was chosen as a first approximation 'cut-off point'. This leads chamber temperatures of less than 1800 K as acceptable.

Figure 3 7 shows the insulation liner, the temperature from T3 to T4, is the most significant component to reduce the chamber wall temperature. The thickness of said liner allows for higher operating temperatures, therefore combustion performance. However, assuming thickness of the other materials are constant, a, increase in liner thickness reduces the combustion chamber radius therefore reduce volume of fuel. However, it could increase specific impulse as combustion can occur at an O/F to achieve a high C*. As mentioned earlier, this will be the focus of the fuel work package after the PDR.

As mentioned earlier, these equations assume steady state heat transfer. This is a significant limitation as the short burn duration may never see steady state conditions. In reality, the heat transfer may not penetrate the entirety of the material. This causes a significant underestimate of the maximum allowable chamber temperature. Therefore, the future work requires a transient CFD simulation of the heat transfer characteristics in the combustion chamber.

Moreover, the uniform heat transfer was assumed throughout the material. For HDPE this assumption is valid however paraffin is prone to voids while casting which can cause cracks and causes heat spots across the combustion chamber. Furthermore, O/F changes over time, therefore the chamber temperatures decrease over time. A transient CFD will be carried out with the grain shape and dimensions of the combustion chamber modelled over time to obtain more accurate chamber temperature, chamber wall relationship.

Material Selection and Sizing for Each Layer

1. **Fuel**: The fuel grain plays a crucial role in absorbing significant amounts of heat due to its high specific heat while limiting heat transfer due to its low thermal conductivity. This characteristic allows a thick layer of fuel to not only provide the necessary fuel for combustion but also act as a heat sink, effectively mitigating temperature rise in the system. For our application, High-Density Polyethylene (HDPE) with a thermal conductivity of 0.49 W/mK and paraffin wax with a thermal conductivity of 7.1 W/mK were considered. Based on heat transfer performance, HDPE is the preferred choice for the fuel grain. The fuel grain thickness has been selected as 2 cm to optimally utilize the low thermal conductivity of HDPE while ensuring adequate fuel supply for the combustion process.
2. **Insulation**: A variety of insulation materials are available, but for our system, the focus is on materials with low thermal conductivity values. Cork, with a thermal conductivity of 0.036 W/mK, has been identified as the ideal insulation material due to its excellent heat resistance and cost-effectiveness. To maximize the insulation's effectiveness, it should ideally be as thick as possible. However, space and weight constraints must be carefully considered, and the final decision on thickness will be made in consultation with the Mechanics team to balance thermal performance and system design limitations.
3. **Metal Casing:** The material selected for the combustion chamber wall must balance thermal conductivity with the ability to maintain structural integrity under high temperatures. Copper alloys have been chosen as the preferred material due to their relatively high thermal conductivity (162 W/mK) and their ability to retain stiffness at elevated temperatures. While aluminum offers higher thermal conductivity (237 W/mK), it has a lower melting point and loses its stiffness at

approximately one-third of its melting point, making it unsuitable for high-temperature applications where structural integrity is critical. Copper alloys, on the other hand, provide a better combination of heat management and mechanical strength at high temperatures, making them the optimal choice for the combustion chamber wall. The thickness of the metal casing has been set at 7.1mm to balance effective heat transfer and maintain the necessary structural strength to withstand the high hoop stresses within the chamber.

The calculation is based on Electrical Analogy in Steady State Heat Transfer Analysis which requires:

Heat,in = Heat, out

Q,con1 = Q,cond1 = Q,cond2 = Q,cond3 = Q,con2 = Q = (Ta1 – Ta2)/R,tot

R,tot = R,con1 + R,cond1 + R,cond2 + R,cond3 + R, con2

Q = (Ta1 – T1)/R,con1

- ð T1 = Ta1 – Q*R,con1
- ð T2 = T1 – Q*R,cond1
- ð T3 = T2 – Q*R,cond2
- ð T4 = T3 – Q*R,con2

Where,

Q,con1 = Convection heat transfer from the core [W/m^2K]

Q,cond1 = Conduction heat transfer through the fuel grain [W/mK]

Q,cond2 = Conduction heat transfer through the insulation [W/mK]

Q,cond3 = Conduction heat transfer through the metal [W/mK]

Q,con2 = Convection heat transfer to ambient [W/m^2K]

R,con1 = Convection resistance from core

R,cond1 = Conduction resistance by fuel grain

R,cond2 = Convection resistance by insulation

R,cond3 = Convection resistance by metal casing

R,con1 = Convection resistance from the ambient air

Calculation of internal convection heat transfer coefficient h, internal

h,internal = C *(Cp,mix*G^0.8)/Di^0.2*(1+(Di/L)^0.7)----------[ref]

This equation show that convection coefficient varies axially and is maximum at the end of chamber. Thus, our calculations will be based on the throat of nozzle.

Cp,mix = specific heat of mixture

S = cross-sectional area

W = mass flow rate = wp/tp

Wp = mass of propellent kg

tp = burn time ses

G = mass velocity = W/S

Di = chamber inside diameter

Calculation of external heat transfer coefficient h,external

Nu = h2*D/k = 0.3 + (0.06*Re^1/2*Pr^1/3)/(1 + (0.4/Pr)^2/3)^1/4 * (1 + (Re/282000)^5/8)^4/5 ----------[ref]

Nu = Nusselt's number

Heat transfer coefficient external = h,external [W/m^2K]

D = External diameter

K = thermal conductivity of air [w/mK]

Re = Reynolds Number

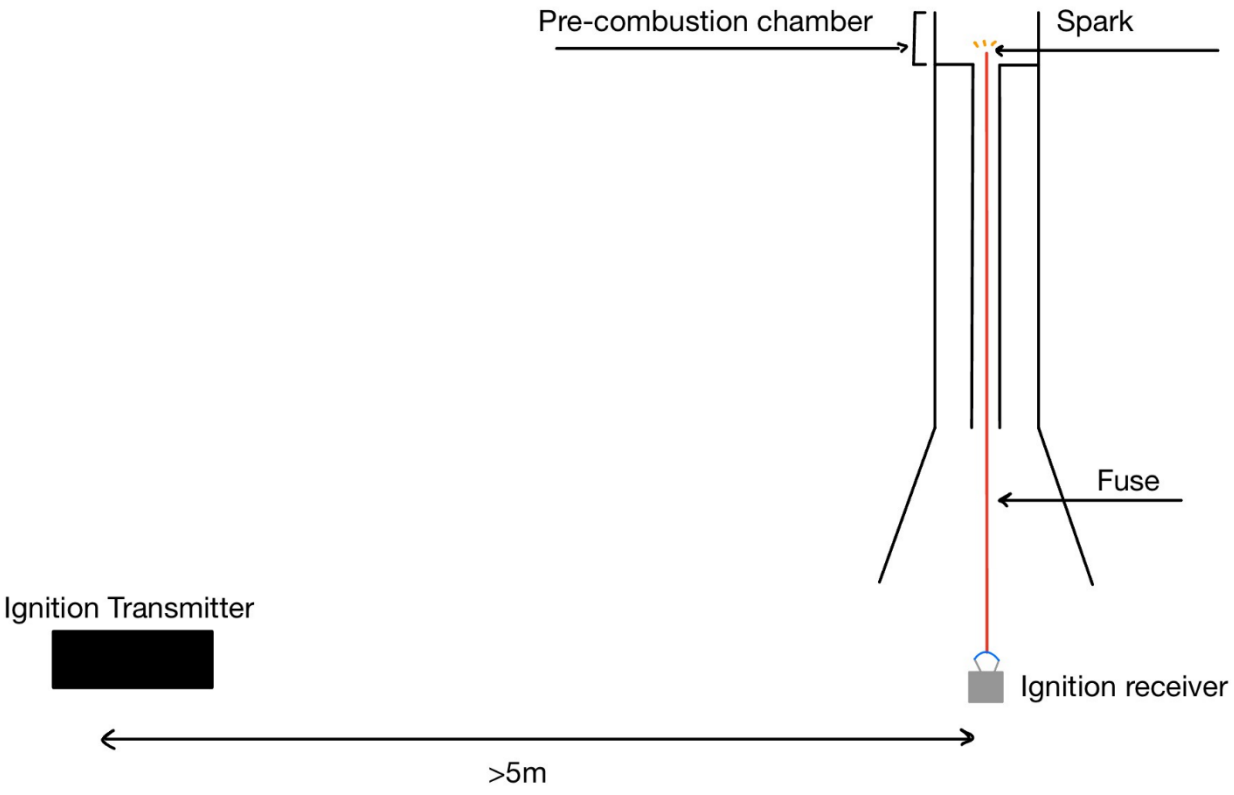
Pr = Pradntl number

Thermophysical Properties and Chamber Sizing

- **Core Thickness** (r1): 0.01 cm
- **Fuel Grain:** thickness **=** 0.02 cm and thermal conductivity = 0.49 W/mK
- **Insulation**: thickness = 0.05 cm and thermal conductivity = 0.036 W/mK
- **Metal Casing**: thickness = 0.08 cm and thermal conductivity = 162 W/mK

5.8 Ignition System

The ignition system will consist of a signal transmitter, a signal receiver, a 15kV arc generator and a fuse . The transmitter is responsible for the start-up and shutdown of the ignition system and must be placed at least five meters away from the rocket to comply with safety regulations. The receiver will be the arc generator connected to an Arduino kit in a protective casing, the fuse will be placed inside the receiver casing in between the two prongs of the arc generator where it will be ignited by the generated arc on start-up. The fuse will then travel up the combustion chamber into the pre-combustion chamber where it will ignite the oxidiser. The assumption now is that the pre-combustion chamber will need to be one hundred percent saturated with oxidiser to ensure ignition, however this is subject to change.



5.8.1 Ignitor-Electronics

5.8.2 Define the type of igniter will be used.

15kV DC High Voltage Arc Ignitor From Combustion team

5.8.3 Define type of components and hardware needed for an ignition system.

Category	Component/Hardware	Specification/Details
Power Supply	Step-up Transformer	15kV output, suitable for input voltage of 12-24V DC
	Capacitors	High-voltage capacitors (15kV, 10-100nF)
	Diodes/Rectifiers	High-voltage diodes for AC to DC conversion
Control Circuit	Microcontroller/Control Board	Arduino, Raspberry Pi, or custom controller with TTL (5V logic)
	Optocouplers	High-voltage optocouplers for isolation
	Triggering Relay/Solid-State Relay (SSR)	High-voltage rated relay (15kV or more)
	Protection Circuitry	Fuses, MOVs (Metal Oxide Varistors)

Switching Components	Arc Gap/Electrodes	Spark plug or high-voltage ignition electrode (15kV rated)
	High-voltage Wires and Connectors	Silicone or Teflon insulated cables rated for 15kV DC
Sensors & Feedback	Voltage/Current Sensors	Hall effect sensors, voltage dividers, rated for high-voltage monitoring (15kV)
	Indicator LEDs/Alarms	Green (armed), Red (fault), Yellow (active)
Safety & Enclosure	Insulated Enclosures	High-voltage insulated enclosures, rated for 20kV or higher
	Grounding and Shielding	Grounding terminals, shielding materials
Testing & Diagnostics	Oscilloscope/Multimeter	Digital oscilloscope with high-voltage probes
	High-Voltage Test Equipment	Insulation resistance testers, high-voltage DC test equipment (20kV rated)

5.8.4 System Schematic

- **Power Supply**
  - **Component**: Step-up transformer, capacitors

    - **Functionality**: Converts 12-24V DC input into 15kV DC. Stores energy to generate the arc for ignition.
- **Ignition Control Circuit**
    - **Component**: Microcontroller (Arduino/Raspberry Pi), optocouplers
    - **Functionality**: Manages ignition timing, triggers the arc, and isolates low-voltage control circuitry from the high-voltage system.
- **Triggering Relay (SSR)**
    - **Component**: Solid-state relay (SSR) or mechanical relay
    - **Functionality**: Switches the high-voltage system based on the signal from the microcontroller.
- **High-Voltage Capacitors**
    - **Component**: Capacitors (rated for 15kV)
    - **Functionality**: Stores and releases energy to create a high-voltage arc for ignition.
- **Arc Gap**
    - **Component**: Spark plug or high-voltage electrodes
    - **Functionality**: Forms the gap where the high-voltage arc will initiate.
- **High-Voltage Cables**
    - **Component**: Insulated cables (e.g., silicone, Teflon)
    - **Functionality**: Carries the 15kV DC power from the supply to the arc gap and other components.
- **High-Voltage Diodes**
    - **Component**: Rectifiers (diodes)
    - **Functionality**: Converts AC power input into stable DC to power the ignition system.
- **Voltage/Current Sensors**
    - **Component**: Hall effect sensors, voltage dividers
    - **Functionality**: Monitors voltage and current during ignition to ensure proper operation and provide diagnostic data.
- **Protection Circuit**
    - **Component**: Fuses, MOVs (Metal Oxide Varistors)
    - **Functionality**: Protects the system from overcurrent, voltage spikes, and component failures.
- **Indicator LEDs**
    - **Component**: LEDs
    - **Functionality**: Indicates the status of the ignition system (armed, active, fault conditions).
- **Enclosure & Grounding**
    - **Component**: Insulated housing, grounding terminals

- **Functionality**: Safely houses high-voltage components and prevents accidental electric shock by providing proper grounding.

- **Testing Equipment**
  - **Component**: Oscilloscope, multimeter
  - **Functionality**: Used to monitor voltage, current, and performance during system testing and diagnostics.

5. System Schematic Flow

Input

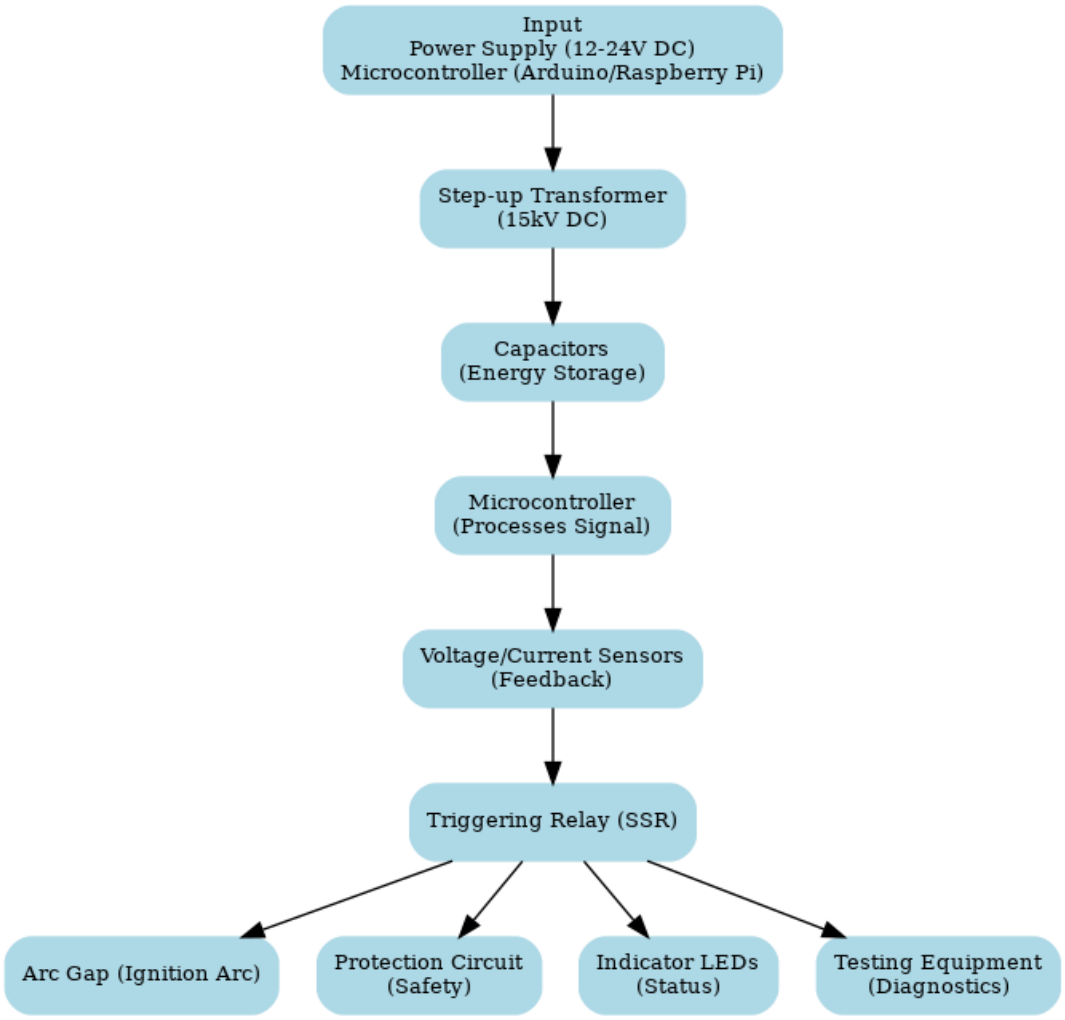
- **Power Supply**: 12-24V DC is provided to the system.
  - **Microcontroller (Arduino/Raspberry Pi)**: Receives commands or input for ignition timing and triggering from the operator or control system.

Processing

- **Step-up Transformer**: Increases the voltage to 15kV DC.
- **Capacitors**: Store the energy needed for the arc ignition.
- **Microcontroller**: Processes the ignition signal, decides when to trigger the arc, and sends the command to the triggering relay.
- **Voltage/Current Sensors**: Monitor the voltage and current for real-time feedback and safety purposes.

Output

- **Triggering Relay (SSR)**: Activates the high-voltage circuit when the microcontroller signals.
- **Arc Gap**: High-voltage arc is created at the spark plug/electrodes, igniting the fuel or propellant.
- **Protection Circuit**: Cuts off the power if any abnormal conditions (e.g., overcurrent or high voltage) are detected.
- **Indicator LEDs**: Display system status (armed, active, fault conditions).
- **Testing Equipment**: Provides diagnostics and performance monitoring via oscilloscope and multimeter.


Input
Power Supply (12-24V DC)
Microcontroller (Arduino/Raspberry Pi)
Step-up Transformer
(15kV DC)
Capacitors
(Energy Storage)
Microcontroller
(Processes Signal)
Voltage/Current Sensors
(Feedback)
Triggering Relay (SSR)
Arc Gap (Ignition Arc)
Protection Circuit
(Safety)
Indicator LEDs
(Status)
Testing Equipment
(Diagnostics)

5.9 Electronic System (Control, DAQ and HMI)

5.9.1 Control System

5.9.1.1 Introduction

This section focuses on the preliminary design of the control system for the CHURRO engine. It will cover the design methodology, key requirements, system schematics, control system block diagrams, processes, and Simulink models.

The control system for the CHURRO engine consists of three main phases or modes:

1. Start-Up

2. Normal Operation

3. Shut Down

The primary roles of the control system are to ensure a safe, reliable, and efficient start-up and shutdown and to regulate and control the thrust and chamber pressure to achieve the desired performance.

5.9.1.2 Methodology

The methodology utilized to design, develop, and identify the control system for the CHURRO engine is illustrated in Figure 8. This approach is based on the framework outlined in (Dorf & Bishop, 2017, p. 20). It was selected for its simplicity, ease of use, and modernity, making it suitable for the current preliminary design stage. However, it is important to note that this methodology may evolve throughout the project into a more detailed approach, such as the one described by (Huzel & Huang, 1992, p. 226).

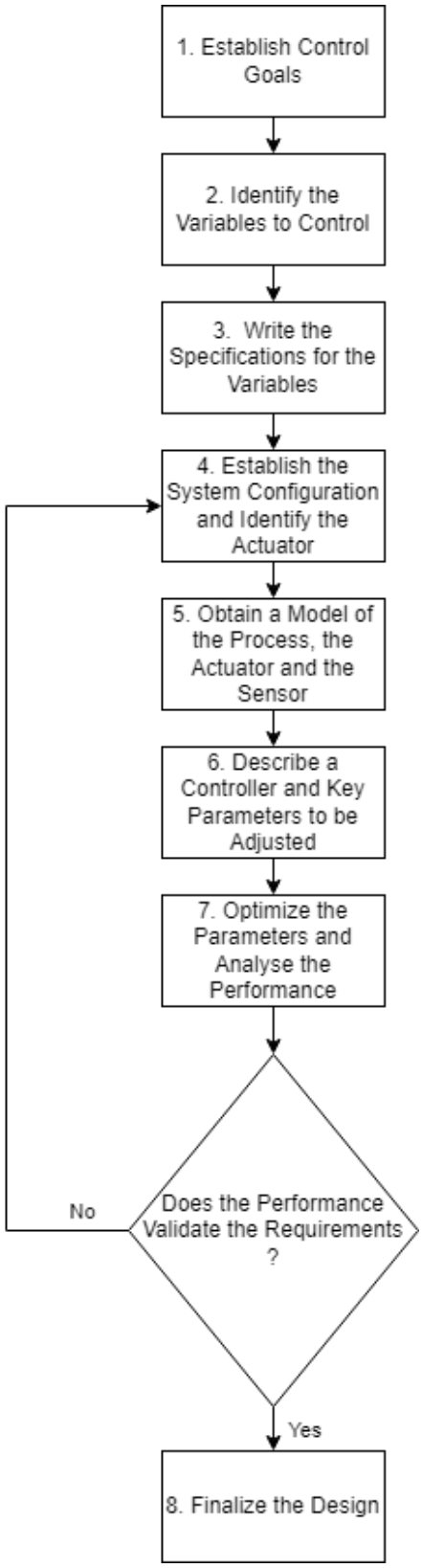


Figure 8 - Control System Design Process

5.9.1.3 Key Requirements

Table 6 outlines the key requirements for the control system. While many of the verification values are still undetermined, it is essential that these values are established in order to verify the system requirements and meet the desired goals and objectives of the control system.

5.9.1.4 Control System Top-level Design

Creating a top-level system schematic is crucial for establishing the system configuration. Figure 9 illustrates the top-level system schematic, highlighting the position of the pressure sensor at the mixing chamber. This sensor will provide the controller with input data regarding the chamber pressure. In turn, the controller will output control commands to the control valve, which regulates the flow of the oxidizer. This can be seen represented by a block diagram in Figure 10. Additionally, tank pressure is another input to monitor the oxidizer tank's pressure; the pressure relief valve will open and release excess pressure if it is over pressured.

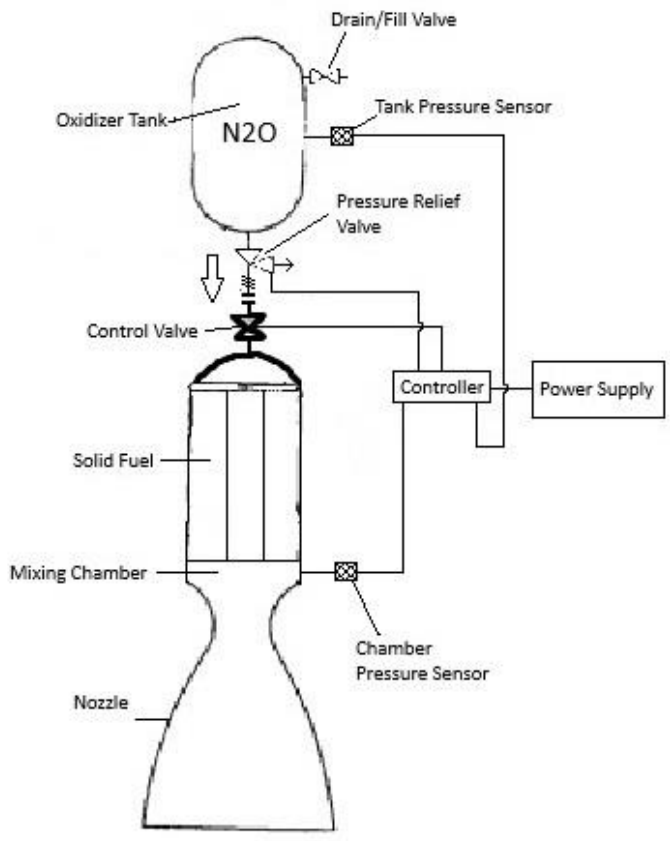


Figure 9 - Top-level Control System Schematic

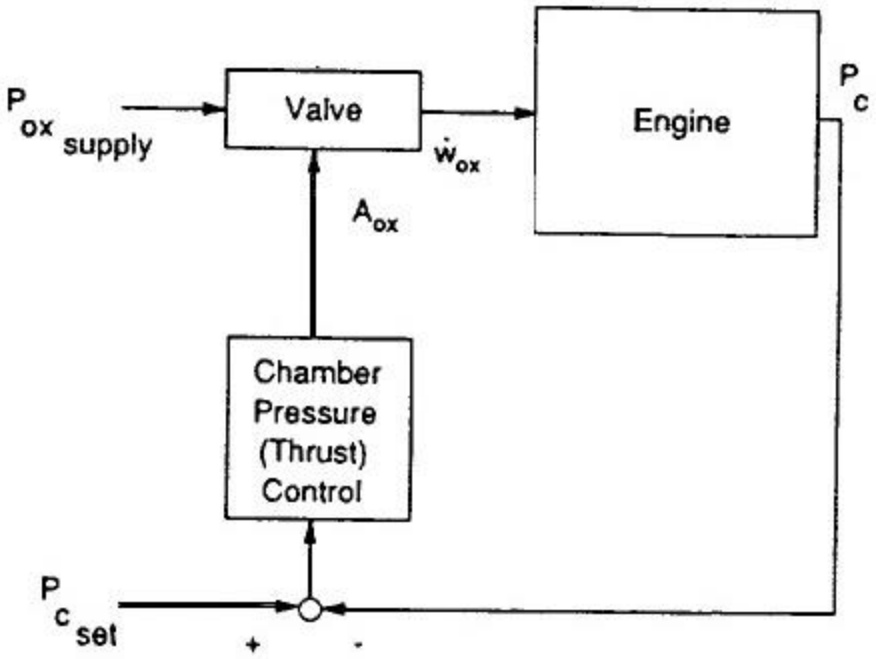


Figure 10 - Chamber Pressure Control Block Diagram

Figure 11 illustrates the control system process for the CHURRO engine. The process begins with purging the system and eliminating residual fluids or objects. This step also serves as a final safety check for potential leaks, faulty sensors, or actuators before starting the engine. Once the purging is complete and no issues are identified, the start-up sequence can commence. During this phase, the purge valve closes, and the oxidizer valve opens to initiate performance and ignition.

During regular operation, the system continuously monitors and regulates the chamber pressure. The control system will automatically shut down the engine if the pressure increases excessively. Conversely, if the system maintains the desired pressure but experiences a loss of pressure, it will adjust the flow valve accordingly. This process continues until the system reaches its specified shutdown conditions or exhausts all oxidizers.

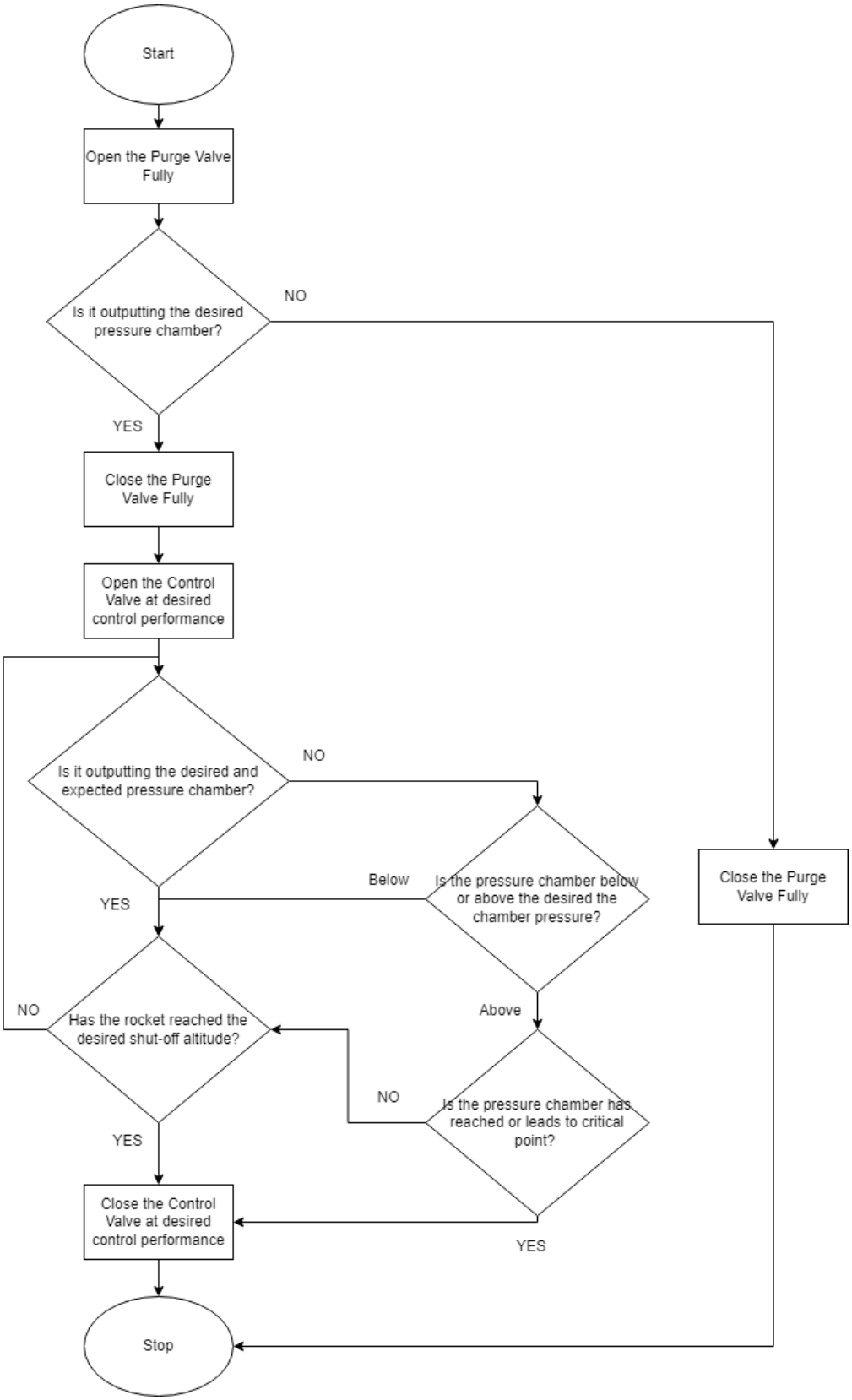


Figure 11 - Control Process Flow Chart

5.9.1.5 Simulation Model

The simulation model for the CHURRO engine's control system, which is manageable and straightforward, was created in Simulink, as shown in Figure 12. This closed-loop system effectively regulates chamber pressure. At this stage, the model is simple yet robust, built on several assumptions, and does not currently account for any disturbances.

The transfer function of the engine model, a key component of our early-stage project, is presented in Equation 1. This transfer function was derived from (Lorenzo & Musgrave, 1992) using parameters from the previous CRANSEDS propulsion team's "DRAKE" hybrid rocket engine. Since we are still in the early stages, not all parameters for the CHURRO have been defined; therefore, we will use the DRAKE parameters to develop the control system model further. The MATLAB code and parameters used to generate the engine transfer model can be found in Appendix C: MATLAB Code.

Equation 1 - Engine Model Transfer Function

$$Engine(s) = e^{-0.004s}\frac{5.663 * 10^{6}}{1.196 * 10^{-8} + 1}$$

The actuator model is a simple one that calculates the flow rate based on the area of the valve's cross-section. The valve area is constrained to the cross-sectional area of the feedlines. Several assumptions have been made regarding the valve, including:

1. Zero pressure drops across the entire system.

2. The pressure in the oxidizer tank is constant.

3. The density of the oxidizer is constant.

4. The cross-sectional area of the valve is equal to that of the feed lines due to using a ball valve.

At this project stage, the control comprises a simple continuous Proportional-Integral (PI) controller without integral windup. The proportional and integral gains are set to the default value of 1. This configuration produces the performance illustrated in Figure 13, with a rapid rise time of 0.006 ms, which may introduce quick temperature spikes, potentially leading to cracking in the combustion chamber. The controller exhibits no overshoot, and the settling time is quick, maintaining a small margin and high accuracy, as defined by an integral square error of 0.002855. It is important to note that the control system model is not finalized and is still in its early development stages, meaning it is subject to significant changes.

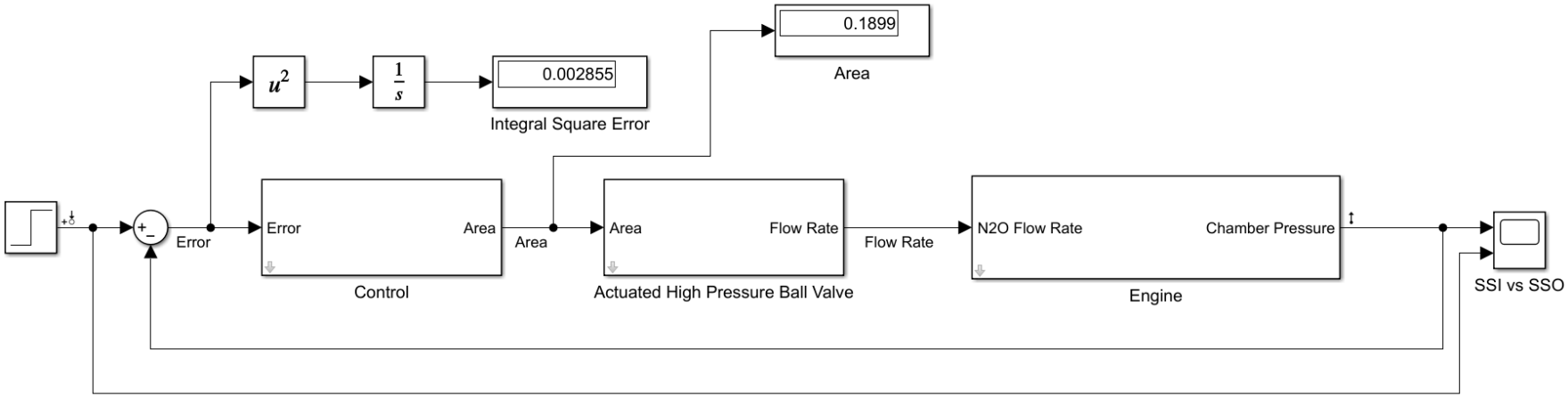


Figure 12 - Control System Simulink Model

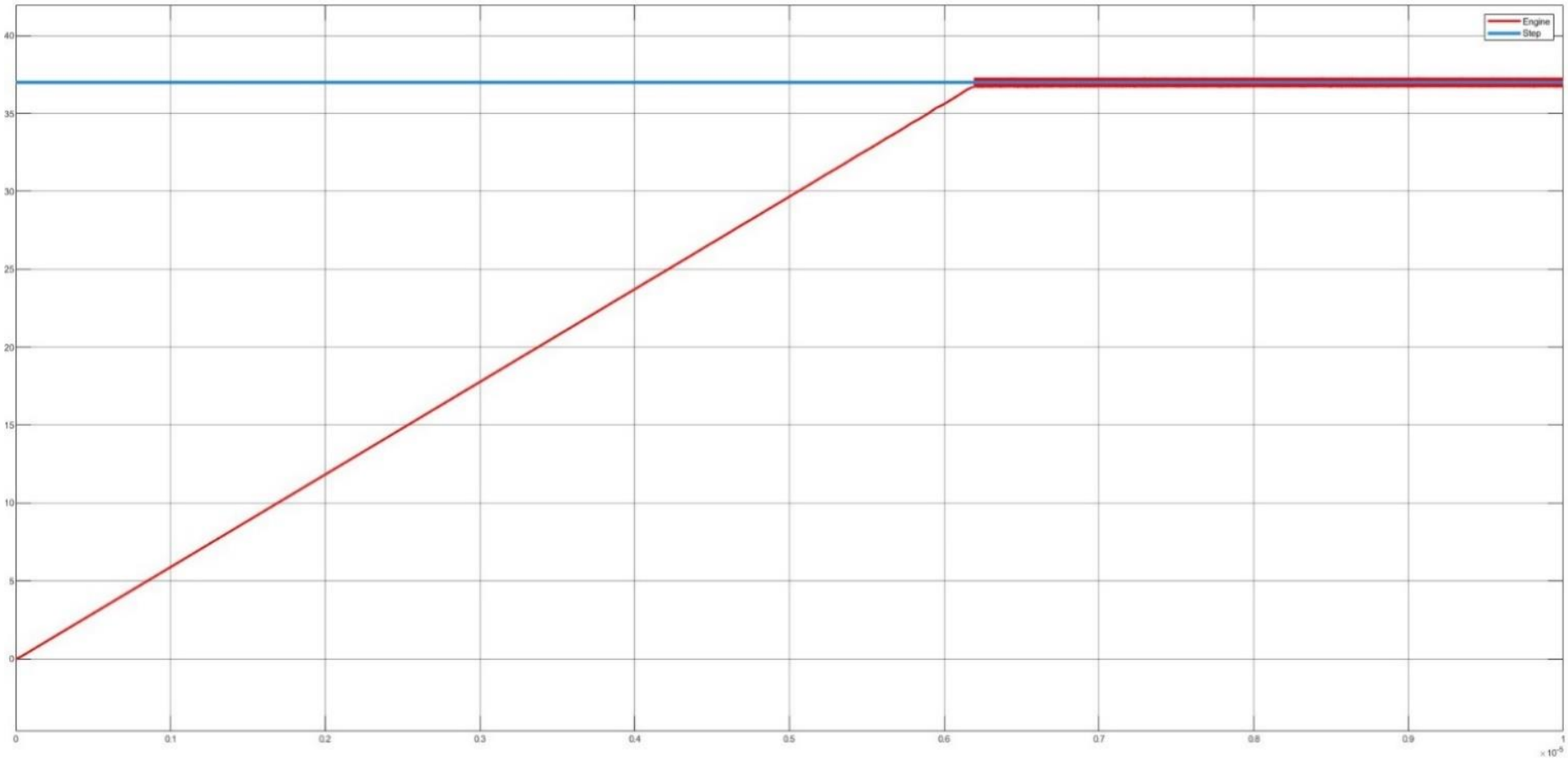


Figure 13 - Control System Performance: SSI vs SSO

5.9.2 DAQ Systems

5.9.2.1 Introduction

This section focuses on the preliminary setup of the Data Acquisition system for the CHURRO engine. It will cover the Data collected, Sensors needed and DAQ System Schematic.

The DAQ System mainly consists of three sensors

1. Load Cell (Thrust Measurement)

The load cell is responsible for measuring the thrust generated by the rocket engine. This sensor converts the mechanical force exerted by the engine into an electrical signal, which is then processed by the DAQ system. The load cell is carefully calibrated to ensure precision and minimize noise, providing reliable data for performance evaluation.

2. Pressure Transducers (Fuel Pressure Ratio Measurement)

Two pressure transducers are used to monitor the pressure in critical locations within the hybrid rocket engine's fuel system. These sensors measure the fuel pressure at the inlet and the chamber, enabling the calculation of the pressure ratio. This data is crucial for assessing the combustion efficiency and ensuring the engine operates within safe pressure limits.

3. Thermocouple Sensors (Temperature Measurement)

Thermocouples are employed to measure the temperature at specific points in the rocket engine, such as the combustion chamber or nozzle exit. These sensors provide real-time thermal data, which is essential for evaluating the engine's thermal performance and identifying potential thermal stress points.

5.9.2.2 DAQ System Architecture

The DAQ system integrates these sensors into a unified framework. The architecture consists of:

1. **Signal Conditioning Units:** Amplify and filter the sensor outputs to ensure noise-free and accurate signals.
2. **Analog-to-Digital Converters (ADCs):** Convert the analog signals from the sensors into digital data for further processing.

3. **Data Logger/Controller:** A central processing unit collects and stores the data, which can be accessed in real time or post-test for analysis. This controller is programmed to handle high sampling rates to capture transient phenomena during engine operation.

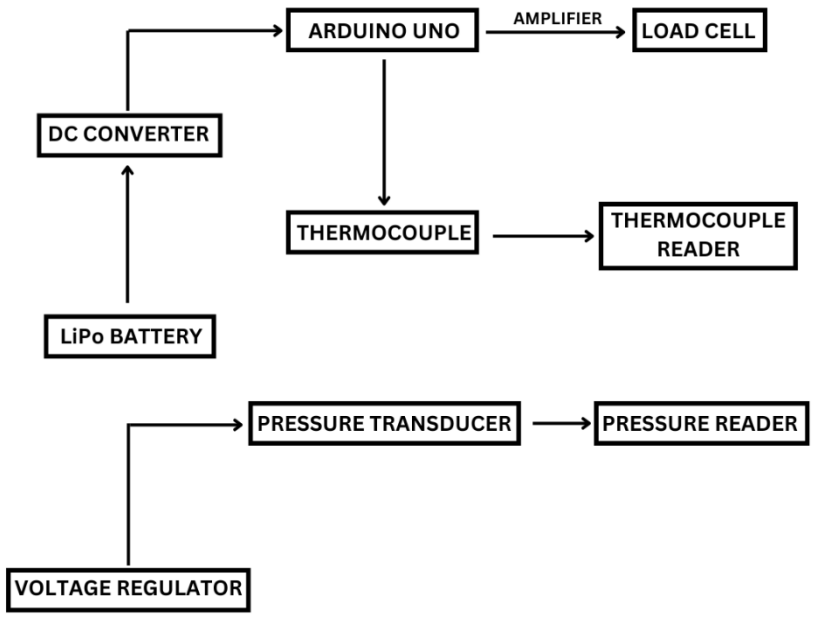


Figure 11 – Block Diagram of DAQ System

5.9.2.3 System Calibration and Validation

Each sensor should be calibrated before integration into the DAQ system using industry-standard procedures. Validation tests will be conducted to ensure the accuracy and reliability of the acquired data under simulated engine conditions.

5.9.2.4 Data Utilization

The acquired data will be used to:

- Evaluate the thrust-to-weight ratio of the engine.
- Monitor and optimize the combustion process.
- Ensure safety by detecting anomalies in pressure and temperature readings.

5.9.3 HMI Systems

5.9.3.1 Top Level Requirements

ID	Requirement	Verification
TLR-01	The HMI system shall display real-time data for pressure, temperature, and fuel flow.	Test the display to ensure accurate real-time updates from sensors.
TLR-02	The HMI system shall be powered by a 5V power source (USB or battery).	Verify power delivery and ensure stable operation under load.
TLR-03	The system shall support communication via Wi-Fi and/or Bluetooth.	Test connectivity and data transfer using ESP8266/HC-05 modules.
TLR-04	The system shall be designed with low-cost components.	Compare the bill of materials (BOM) to budget constraints.
TLR-05	The system shall include a user interface for real-time data display.	Verify the LCD/TFT/OLED display functionality.

5.9.3.2 Define what need Real Time Monitoring(RTM)

Feature	Application
Data Display	Live plots or gauges for thrust vs. time during a static fire test.
Interactive Controls	Activating or deactivating a fuel valve during a test via touchscreen controls.

Alerts and Alarms	Alarm triggers if chamber pressure exceeds safe limits, with a red warning displayed on the screen.
Data Logging	Log thrust and pressure readings to evaluate engine performance post-test.
Remote Monitoring	Observing live data remotely from a safe distance during hazardous engine tests.
User Interface Elements	Gauges, bar charts, line graphs, and toggle switches on a touchscreen display.

5.9.3.3 Hardware and Components Needed

Category	Component	Description	Purpose
Display System	**TFT or LCD Display**	A small **2.8" TFT** or **16x2 LCD** (I2C version recommended) for displaying real-time data.	Displays pressure, temperature, and fuel flow data.
	OLED Display (0.96")	Compact OLED display for showing detailed data in a small form factor.	Displays temperature, pressure, and other metrics.

Sensors and Data Acquisition	**Pressure Sensor (BMP180/MPX5700)**	Low-cost sensor for measuring chamber pressure or other critical areas.	Monitors engine pressure.
	Temperature Sensor (DS18B20)	Digital temperature sensor that works well with Arduino for monitoring engine temperature.	Monitors engine temperature.
	Flow Sensor (YF-S201)	Basic flow sensor used to measure fuel and oxidizer flow rates.	Monitors fuel and oxidizer flow rates.
	Accelerometer (MPU6050)	Low-cost accelerometer for detecting vibrations and acceleration.	Monitors engine stability and performance.
Communication and Interface	**Wi-Fi Module (ESP8266)**	Low-cost Wi-Fi module for wireless data transmission to a ground station or remote monitor.	Transmits sensor data wirelessly.

	Bluetooth Module (HC-05)	Bluetooth module for short-range wireless communication with a smartphone or control device.	Enables wireless monitoring and control.
Power Supply and Distribution	**5V Power Supply (USB/LiPo Battery)**	A simple 5V power source, like a USB adapter or rechargeable LiPo battery.	Powers the Arduino and connected components.
	Power Distribution Board	Power distribution board to manage power between components, ensuring stable operation.	Ensures stable power supply for the system.
Miscellaneous	**Breadboard and Jumper Wires**	For prototyping and connecting all components together.	Makes it easy to build and test circuits.
	Resistors, Capacitors, Connectors	Essential passive components for building circuits and ensuring stable signals.	Ensures proper functioning of sensors and displays.

5.9.3.4 System Schematic

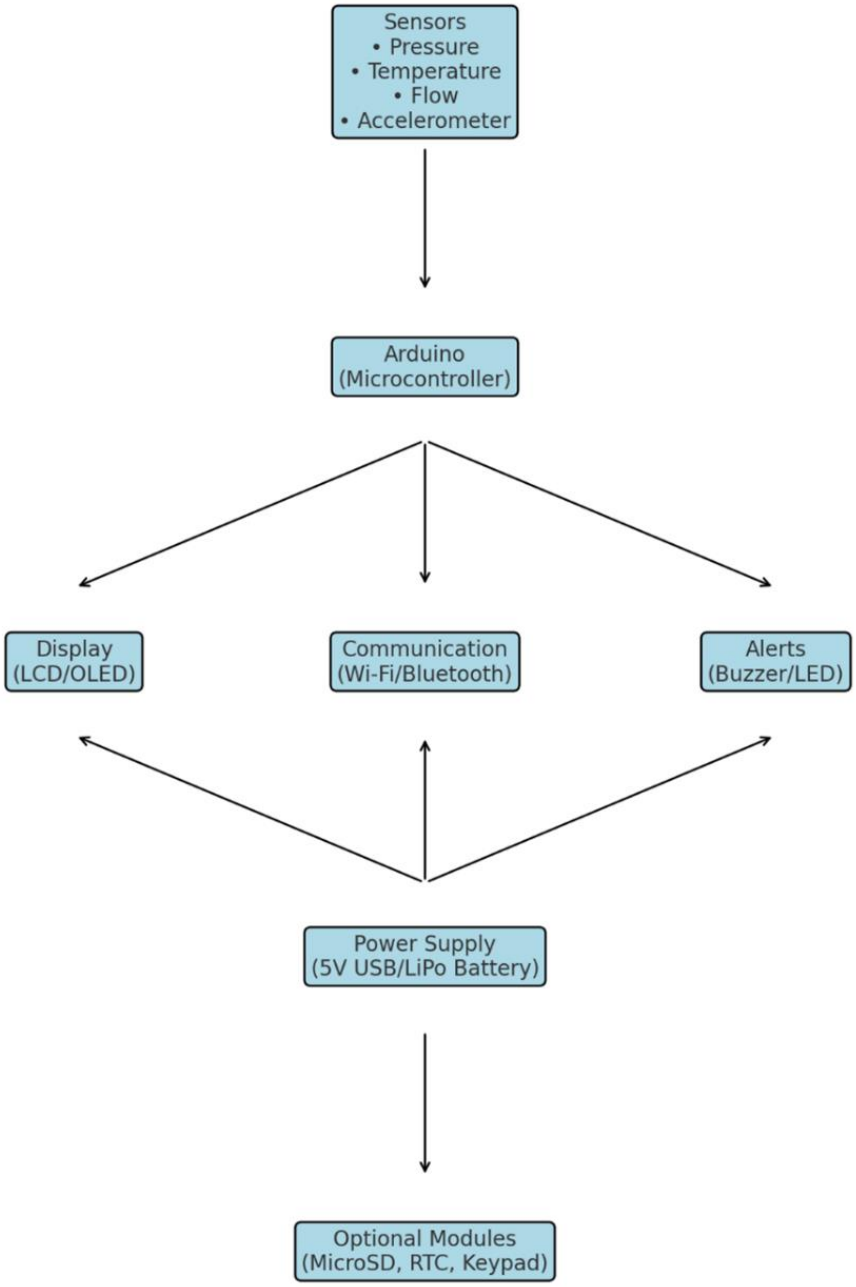
1. **Arduino (Microcontroller)**
    - The central processing unit that collects, processes, and displays sensor data.
    - Interfaces with sensors, display, communication modules, and alerts.
2. **Sensors**
    - **Pressure Sensor**: Measures engine chamber pressure.
    - **Temperature Sensor**: Monitors engine temperature in real-time.
    - **Flow Sensor**: Tracks fuel/oxidizer flow rates.
    - **Accelerometer**: Detects vibrations and anomalies during operation.
3. **Display Unit**
    - **LCD/OLED/TFT Display**: Outputs real-time metrics (e.g., pressure, temperature, flow rate) for user monitoring.
4. **Communication Module**
    - **Wi-Fi (ESP8266)**: Sends sensor data to a remote ground station for monitoring and logging.
    - **Bluetooth (HC-05)**: Enables local wireless monitoring and interaction via smartphone or laptop.
5. **Power Supply**
    - **5V USB/LiPo Battery**: Powers the Arduino and connected components.
    - Includes optional **capacitors** for power stability and **backup battery** for system reliability.
6. **Alert System** (Optional)
    - **Buzzer or LEDs**: Provides alarms for critical conditions like overheating or pressure spikes.
7. **User Input** (Optional)
    - Buttons or rotary encoders for manual control or parameter adjustments.
    - Keypad for advanced input options, like setting thresholds.
8. **Optional Expansion**
    - **MicroSD Card Module**: Logs real-time data for post-test analysis.
    - **RTC Module**: Timestamps data for accurate tracking.

5. System Schematic Flow

1. **Input**
   Sensors (Pressure, Temperature, Flow, Accelerometer) provide real-time data to the Arduino.
2. **Processing**
   The Arduino processes the data and decides:
    - What to display on the screen.

- Whether to send alerts or log data.
- Whether to transmit data wirelessly.

3. **Output**
    - Real-time metrics are shown on the display.
    - Alerts (buzzer/LED) are triggered for critical conditions.
    - Data is transmitted via Wi-Fi/Bluetooth for remote monitoring.



5.10 Oxidiser Filling System

As part of the launch procedure, the oxidiser is loaded onto the vehicle moments before flight. Due to the nature of high-pressure fluid transfer this must be done in an automated manner. The fuelling system has been derived from an existing system operated by Half

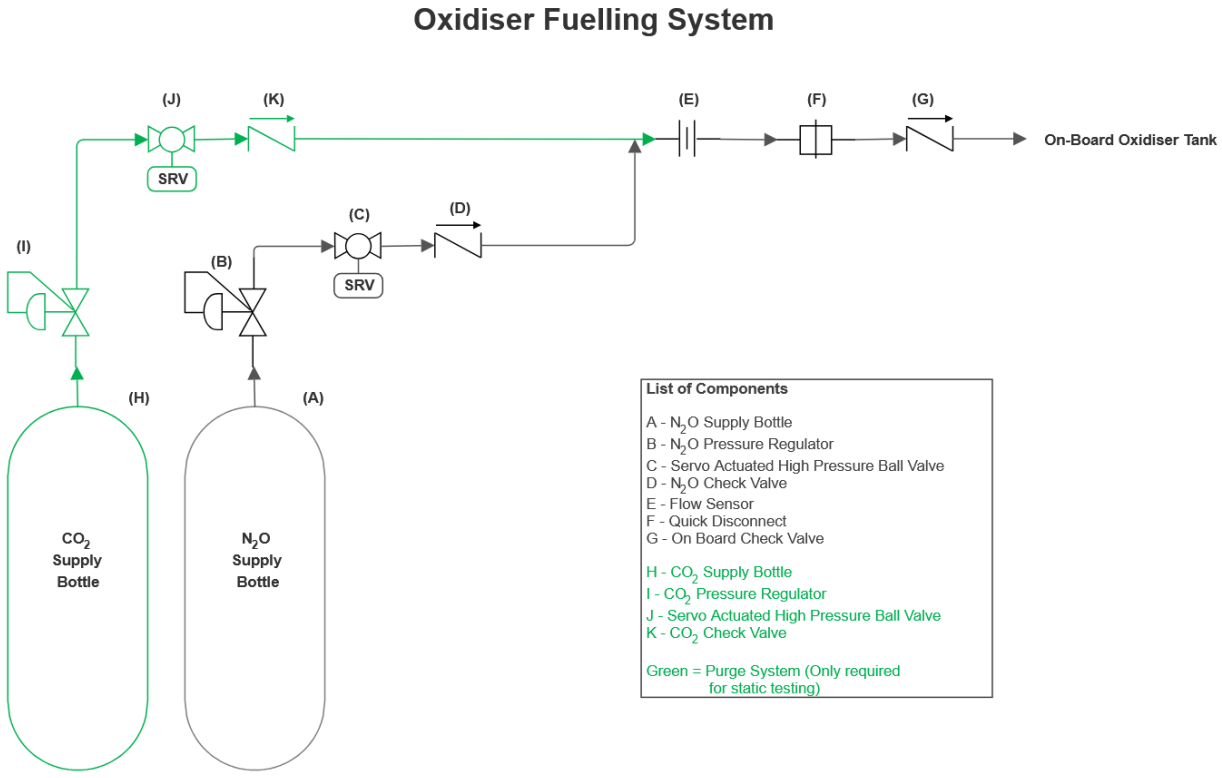
Cat Rocketry for their bipropellant rocket the Mojave Sphinx. Our decision to base the ground equipment off an existing design is mostly due to the short development time frame and testing availability. The Half Cat design is flight proven with excellent resources for us to build upon and modify to suit our needs.

As you can see from the diagram below, there is a nitrous oxide supply line and an optional carbon dioxide line for use during static testing. Both consist of a supply bottle, a pressure regulator, a servo-actuated ball valve and a check valve. These lines feed into a flow sensor, a quick disconnect mounted to the vehicle, a check valve and finally into the oxidiser tank.

Due to the vapour pressure of nitrous oxide, the on-board oxidiser tank will become pressurised quickly. This is undesirable as we need to fill the tank with liquid nitrous oxide, aiming for an ullage of 5-10%. As such a static vent is included at the top of the on-board oxidiser tank, which allows us to keep filling the tank whilst allowing the vapour to escape. The static vent also acts as a safety feature if our depressurisation mechanisms fail, given enough time the tank will naturally depressurise.

The filling procedure shall go as follows:

- Ground system is connected to the vehicle quick disconnect
- The ball valve positions are checked to ensure they are closed
- The supply bottle regulator is opened by hand
- All personnel retreat to a safe distance
- The nitrous oxide line ball valve is opened remotely
- Nitrous oxide begins to flow into the oxidiser tank, some vapour will be observed expelling out of the static vent
- Within 2 minutes the static vent vapour will become denser, this indicates that the liquid is being venting and hence the tank has reached the required fill level for launch (it should be possible to calculate an exact fill time based off the nitrous oxide flow rate however the static vent is a good secondary indicator)
- The nitrous oxide ball valve is closed remotely
- Launch sequence begins, must be completed within 30s otherwise the nitrous oxide line must be reopened to fill the tank and the process redone
- For static testing the system can be purged by opening the carbon dioxide line



5.11 Mechanics and Thermal Analysis

5.11.1 Objective

1. Validate the mechanical integrity and thermal performance of the combustion chamber, nozzle, and oxidizer tank.

2. Ensure all components meet operational conditions, including:

- High internal pressures.
- Elevated temperatures during operation.
- Material properties under extreme thermal and mechanical stress.

5.11.2 Design Overview and Analysis

The design of the combustion chamber, nozzle and oxidizer tank prioritizes structural integrity and thermal resilience under operational conditions. Using CAD models, key dimensions and materials were optimized for high temperatures and pressures. Analytical calculations and FEA simulations validated the components, ensuring compliance with safety and performance requirements.

5.11.2.1 Combustion Chamber

Design Parameters for the Combustion Chamber

- Internal Radius (r): 24.5 mm.
- Wall Thickness (t): 7.01 mm.
- Length: As per operational requirements (not provided in the data but must be optimized for nozzle attachment).
- Material Properties:
- Material: Copper Alloy/ Stainless Steel (TBC)
- Density: 2.70 g/cm³.
- Yield Strength: ~240 MPa.
- Ultimate Tensile Strength: ~290 MPa.
- Thermal Conductivity: 166 W/m·K.
- Melting Point: ~650°C.
- Thermal Expansion: 23.4×10−6/K23.4×10−6/K.
- Operating Conditions:
- Design Pressure (pp): 4.6 MPa.
- Operating Temperature: Based on fuel/oxidizer combination (e.g., N2O and fuel).
- Volume: Defined by internal radius and chamber length.

Stress Analysis:

The provided calculations yield:

Hoop Stress (σ_{hoop}):

$$\sigma h = p.rt \quad \sigma h = p.rt$$
$$= 19.3 MPa.$$

Axial Stress (σ_{axial}):

$$\sigma a = \sigma h2 \quad \sigma a = \sigma h2$$
$$= 9.7 MPa.$$

Shear Stress (σ_{shear}):

$$\sigma s = \sigma a2 \quad \sigma s = \sigma a2$$
$$= 4.9 MPa$$

Von Mises Equivalent Stress ($\sigma total$ $\sigma total$)
Using the Von Mises criterion:

$$\sigma total = \sqrt{\sigma h2 + \sigma a2 + \sigma s2} \quad \sigma total = \sqrt{\sigma h2 + \sigma a2 + \sigma s2}$$
$$+ 22.14 MPa.$$

5.11.2.2 Nozzle

- Design Parameters:
  Nozzle Geometry: GVR Rao Parabola Nozzle
  Throat Diameter:
  Throat Area:
  Exit Diameter:
  Exit Area:
  Expansion Ratio:
- Stress Analysis (Theoretical):

- Risk Analysis:

5.11.2.3 Oxidiser Tank

- Design Parameters:

  Tank geometry: Cylindrical with spherical ends

  Tank type: Thin-walled

  Material: AL 6061

- Generic physical and mechanical properties of AL 6061:

Density: 2.70 g/cm³	Melting Point: 650 °C
Thermal Expansion: 23.4 x10-6 /K	Thermal Conductivity: 166 W/m.K
Tensile Strength (Yield): 240 MPa	Tensile Strength (Ultimate): 290 MPa

- Mechanical Analysis (Theoretical):

  The following parameters were used in the calculations:

– Internal radius (r): 50 mm
– Thickness (t): 2 mm
– Design pressure (p): 5.5 MPa
– Tank volume (V): 2500 cc

Using the specified parameters, the hoop stress and axial stress were evaluated to assess the structural integrity under the given operating conditions. The hoop stress (σ_{hoop}) was calculated using the formula:

$$\sigma_{hoop} = \frac{p \cdot r}{t}$$

Similarly, the axial stress (σ_{axial}) was determined as:

$$\sigma_{axial} = \frac{p \cdot r}{2t}$$

To assess the combined loading conditions, the Von Mises stress was calculated. The Von Mises criterion is used to predict the yielding of materials under complex loading by combining the effects of normal and shear stresses. In cases where the loading is uniaxial or purely axial (as in this analysis), the Von Mises stress simplifies to the following form:

$$\sigma_{equivalent} = \sqrt{(\sigma_{hoop})^2 - (\sigma_{hoop} \cdot \sigma_{axial}) + (\sigma_{axial})^2}$$

By applying these formulas, the following values have been calculated based on the given parameters:

σ_{hoop}	σ_{axial}	$\sigma_{equivalent}$
$= 137.5\ MPa$	$= 68.75\ MPa$	$= 119.08\ MPa$

These calculations provide the stress within the structure, ensuring compliance with design safety requirements.

- Factor of Safety (FoS):

  The Factor of Safety (FoS) is a critical parameter used to ensure that a structure or component will not fail under expected loading conditions. It is defined as the ratio of the material's strength to the actual applied stress. The Factor of Safety can be calculated for different stresses, including Hoop Stress, Axial Stress, and Von Mises Stress.

FoS for Hoop Stress,

$$FoS_{hoop} = \frac{\sigma_{yield}}{\sigma_{hoop}}$$

FoS for axial Stress,

$$FoS_{axial} = \frac{\sigma_{yield}}{\sigma_{axial}}$$

FoS for equivalent Stress,

$$FoS_{\text{equivalent}} = \frac{\sigma_{yield}}{\sigma_{\text{equivalent}}}$$

By applying these formulas, the following values of Factor of Safety have been calculated:

$$FoS_{hoop} = 1.75 \qquad FoS_{axial} = 3.49 \qquad FoS_{\text{equivalent}} = 2.02$$

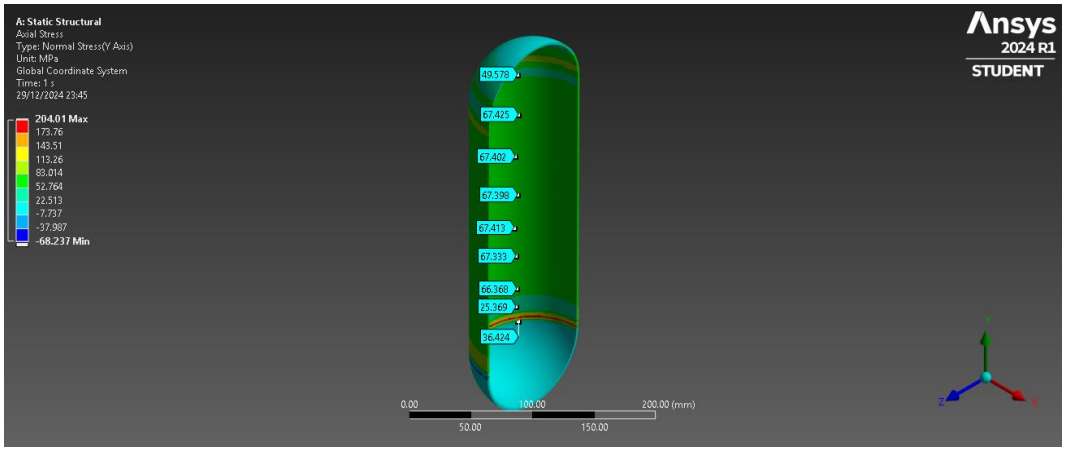
For critical components like the combustion chamber, nozzle and oxidizer tank, a higher FoS may be employed depending on factors such as the consequences of failure, the material properties and the operating conditions to ensure reliable performance and safety under extreme loading conditions.

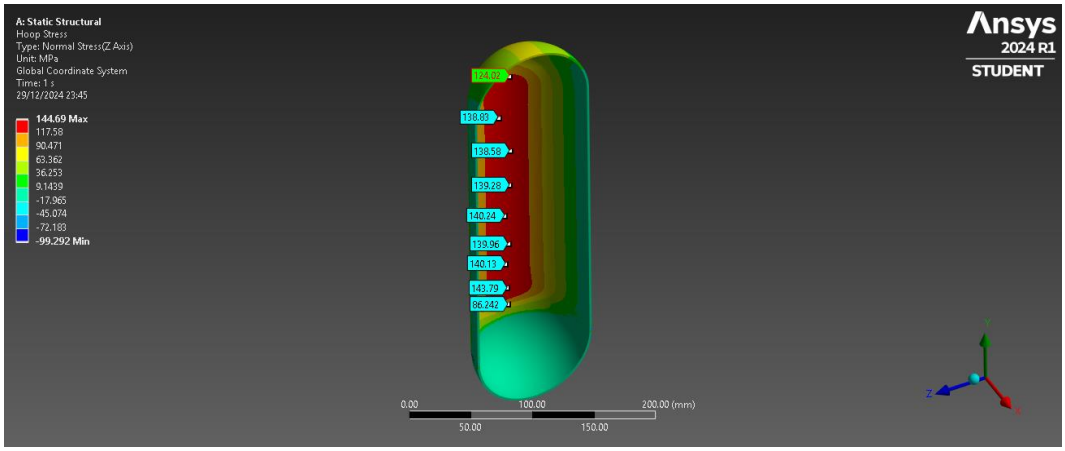
A typical Factor of Safety (FoS) for engineering applications generally ranges from 1.5 to 3.

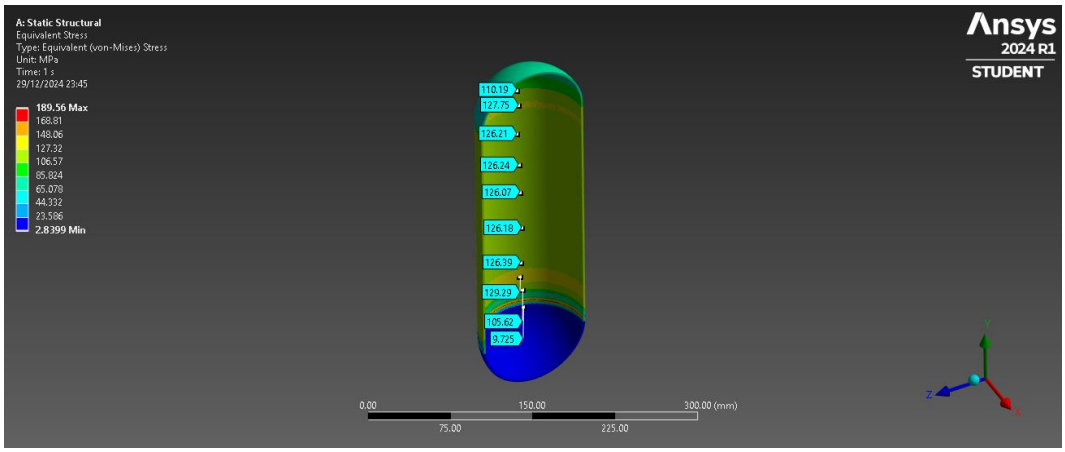
- Finite Element Analysis (FEA):

The next step involves conducting a Finite Element Analysis (FEA) to assess the structural integrity of the oxidizer tank under internal pressure conditions. This analysis will provide detailed insights into the structural behaviour, enabling design optimization and ensuring the components meet the required safety and performance standards and validating the theoretical calculations before physical testing.

The following figures present the results of the FEA simulations,







- Results:

The following table compares the theoretical and FEA results, including the percentage difference and Factor of Safety (FoS) for both theoretical and FEA calculations:

Comparison of Theoretical and FEA Results					
	Results Stress (MPa)		% Difference	Factor of Safety (FoS)	
	Theoretical	FEA		Theoretical	FEA

Hoop Stress	137.5	141.46	2.880	1.75	1.70
Axial Stress	68.75	67.58	-1.702	3.49	3.55
Equivalent	119.08	126.26	6.031	2.02	1.90

- Interpretation and conclusion:

  The comparison between the theoretical calculations and FEA results for hoop stress, axial stress, and equivalent stress shows good agreement, with minimal differences:

  - Interpretation of Results:
    - Hoop Stress: The theoretical value of 137.5 MPa showed a 2.88% difference from the FEA value of 141.46 MPa. The Factor of Safety (FoS) was slightly lower in FEA (1.70) compared to theory (1.75), indicating a marginal reduction in safety under actual conditions.
    - Axial Stress: The difference between theoretical (68.75 MPa) and FEA (67.58 MPa) values was -1.70%, with both FoS values (3.49 for theory, 3.55 for FEA) indicating a safe design.
    - Equivalent Stress: A 6.03% difference was noted between the theoretical (119.08 MPa) and FEA (126.26 MPa) values. The FoS in FEA (1.90) was slightly lower than in theory (2.02), suggesting a higher stress concentration under real conditions.
  - Conclusion:
    - The design of the combustion chamber, nozzle, and oxidizer tank meets the required safety margins, with the Factor of Safety values indicating a robust structure. The minor differences between the theoretical and FEA results fall within acceptable tolerances and confirm that the design can withstand expected loading conditions.
    - Further refinement of the FEA model could improve accuracy, but the current results validate the theoretical calculations and suggest a safe and reliable design.
- Risk Analysis:

The following outlines the identified risks, limitations, and corresponding mitigation strategies for the rocket tank design:

Risks

1. **Material Failures:**
    - **Risk:** The aluminium alloy (AL 6061) may experience material fatigue or failure under cyclic loading or prolonged use, which could lead to catastrophic tank rupture and mission failure.
    - **Mitigation:** Perform extensive fatigue and stress testing to validate AL 6061's suitability under the expected operational conditions. Consider alternative materials with higher yield strength or better corrosion resistance if required.
2. **Stress Exceedance:**
    - **Risk:** Hoop stress (141.46 MPa) and axial stress (67.58 MPa) may approach or exceed the material's yield strength under unexpected pressure or environmental conditions, causing structural deformation or fracture.
    - **Mitigation:** Implement a safety factor in the design to handle stress exceedance. Use finite element analysis (FEA) to optimize stress distribution to ensure that the tank can withstand unanticipated pressure and loading conditions.
3. **Manufacturing Defects:**
    - **Risk:** Imperfections such as cracks, voids, or inclusions during manufacturing could reduce the structural integrity of the tank, leading to potential failure under operational loads.
    - **Mitigation:** Adopt stringent quality control measures during fabrication, including **non-destructive testing (NDT)** methods (e.g., ultrasonic or X-ray inspection) to detect any manufacturing defects before assembly.
4. **Corrosion:**

- **Risk:** Exposure to environmental factors such as moisture and extreme temperatures may lead to corrosion, reducing the material’s strength and durability over time.
- **Mitigation:** Apply protective coatings, such as anodization or corrosion-resistant coatings, to enhance the tank's durability. Regular inspections will be conducted to monitor any corrosion and mitigate it proactively.

5. **Thermal Expansion:**
    - **Risk:** Temperature variations during operation may cause thermal expansion or contraction, creating localized stress concentrations that could lead to material fatigue or failure.
    - **Mitigation:** Implement thermal insulation or cooling systems to manage temperature fluctuations. Perform thermal analysis to predict temperature distribution and assess the impact on material integrity.
6. **Leakage:**
    - **Risk:** Improper sealing or microcracks could lead to leakage of N2O, resulting in loss of fuel or pressurization, thereby compromising the mission.
    - **Mitigation:** Utilize high-quality seals and gaskets in the design. Perform pressure testing of the tank before deployment to ensure it is free from leaks and is adequately sealed.

6 Testing

6.1 Aims of Full-Stack Test

As stated above the main aim of this project is to produce a rocket engine capable of launching CranSEDS EUROC rocket to an altitude of 3000m. Therefore, our testing procedure will need to validate the integration of the motor into the rocket structure, the fill procedure and the performance of the motor itself.

At this early stage of design, detailed procedure does not exist yet as the design of the motor has not yet been finalised. The following section will however outline a procedure based on the preliminary design presented above.

6.2 Current Rocket Design

6.3 General Testing Procedure

6.3.1.1 Checks and Set-up

1. Arrive at test site.
2. Unload equipment and verify that it has arrived and is undamaged. This includes both the motor and airframe.
3. Complete any necessary assembly of motor and rocket such as integration of avionics in airframe and sensor lines on the motor.
4. Insert and attach motor to airframe.
5. Check once again all attachments.

6.4 Control systems

1. Objective

To validate the performance, reliability, and safety of the chamber pressure control system for a hybrid rocket engine under various operating conditions.

---

2. Prerequisites

1. Test Equipment:

- Pressure transducers calibrated to the operating range.
- Data acquisition system (DAQ) for recording pressure and control signals.
- Simulated oxidizer and fuel flow control valves.
- Backup pressure relief systems.

2. Setup:

- Fully assembled rocket engine test stand.
- Functional control system with updated firmware/software.
- Secure communication with ground control and monitoring systems.
- Proper integration of sensors and actuators with the control system.
- Safety systems operational (emergency shut-off, fire suppression).

3. Safety Checks:

- Verify all pressure vessels are within certification limits.
- Ensure all personnel are at a safe distance during testing.
- Conduct pre-test safety briefing.

3.

---

3. Test Steps

3.1. Sensor Calibration

1. Verify sensor calibration against a certified pressure standard.

2. Confirm sensor accuracy to ±1% of the full-scale range.
3. Test sensor response time by simulating rapid pressure changes.

3.2. Actuator Testing

1. Manually actuate the valves (open/close) and verify response.
2. Measure actuator speed and ensure it meets the design specifications.
3. Confirm valve positions are correctly reported to the control system.

3.3. Static Pressure Test

1. Apply a static pressure to the chamber using an external pressurized system.
2. Monitor sensor readings for stability and accuracy.
3. Verify the control system maintains the pressure within ±2% of the setpoint.

3.4. Dynamic Response Test

1. Introduce a step change in desired chamber pressure via the control system interface.
2. Record the time taken for the system to stabilize at the new setpoint.
3. Verify the system remains within ±5% of the desired pressure during transients.

3.5. Feedback Loop Validation

1. Use simulated signals to test the feedback loop functionality:
    - Gradually increase pressure to test overshoot behaviour.
    - Introduce oscillations and observe damping performance.
2. Measure system bandwidth and ensure stability margins are within design parameters.

3.6. Full-Scale Integration Test

1. Conduct a cold flow test (oxidizer only) to ensure system integrity without ignition.
    - Verify the control system adjusts valve positions to maintain setpoint.

- Observe system behavior under simulated oxidizer mass flow rates.

2. Perform a short-duration hot fire test:
    - Start the ignition sequence and monitor chamber pressure in real-time.
    - Record system response to changes in oxidizer and fuel flow rates.
    - Evaluate the system's ability to maintain steady-state operation.

3.7. Fault Injection Tests

1. Simulate sensor failures (e.g., disconnect or provide erroneous data).
2. Observe the system's response to actuator malfunctions (e.g., slow response, failure to close).
3. Verify the system activates safety protocols (e.g., shutting off flow) under fault conditions.

---

4. Pass/Fail Criteria

1. Pressure control must remain within ±5% of the setpoint during steady-state operation.
2. The system must stabilize within 2 seconds after a step change in setpoint.
3. Safety mechanisms must engage within 1 second of detecting a fault.
4. All sensors and actuators must operate within design specifications.

---

5. Data Analysis and Reporting

1. Review recorded data to assess:
    - Transient response characteristics (rise time, settling time, overshoot).
    - Steady-state accuracy.
    - Fault response effectiveness.

2. Generate a test report detailing:
    - Test results with plots of chamber pressure vs. time.
    - Identified issues or deviations.
    - Recommendations for system improvements (if necessary).

---

6. Post-Test Activities

1. Depressurize all systems and secure the test stand.
2. Inspect hardware for damage or wear.
3. Archive test data and update system documentation

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Appendix A: Fuel Selection

Pugh Decision Matrix of candidate fuels

Table 10: Pugh Decision Matrix

Decision Matrix	**Cost**	**Safety/Handling**	**Performance (ignition)**	**Performance (Cstar)**	**Manufacturing**	**Available Data**	**Total**
Prioritise Cost, Safety and Manufacturing	**0.25**	**0.35**	**0.05**	**0.1**	**0.2**	**0.05**	**1**
High Density Polyethelene (HDPE)	8	8	6	7	9	9	8.05
Poly-Methyl Methacrylate (PMMA)	6	9	8	8	8	8	7.85
Poly-Vinyl Chloride (PVC)	8	6	5	6	8	7	6.9
Hydroxyl Terminated Poly-Butadiene (HTPB)	4	7	9	9	6	9	6.45
Acrylonitrile Butadiene Styrene (ABS)	7	8	7	7	9	6	7.7
Parrafin Wax	8	9	8	8	7	9	8.2

Nylon	7	7	6	7	6	7	6.75
Plexiglas (Lucite)	6	9	8	8	8	8	7.85
Polylactic acid (PCA)	6	9	6	6	7	7	7.3
Glycidylazide Polymer (GAP)	2	4	10	10	5	5	4.65
Carboxyl-terminated polybutadiene (CTPB)	4	6	9	9	6	7	6
Ploystyrene (PS)	7	8	5	6	8	6	7.3

Appendix C: MATLAB Code

Engine Model Generation

```
clc

%Engine Model Transfer Function

% Define Constants
c_star =  5277.8; %Characteristic Velocity (m/s)
A_T = 9.5*10^-5; %Throat Area (m^2)
g = 9.81; %Gravitational Constant
sigma = 0.004; %Combustion Delay (s)
e = 2.7182; %Exponential
r_c = 0.02; %Radius of Chamber (m)
L_c = 0.05; %Length of Chamber (m)
A_c = pi *r_c^2; %Area of Chamber (m^2)
V_c = A_c * L_c; %Volume of Chamber (m^3)
R = 8.314; %Gas Constant (J/mol*K)
T_c =  3107; %Combustion Temperature (K)
t = (c_star * A_T * g * V_c)/(R*T_c); %Chamber Fill time (s)
%Create Engine Tranfer Function
s = tf('s');
sys1 = c_star/(A_T*g);
sys2 = exp(-sigma*s)/(t*s+1);
sys3 = series(sys1,sys2);

%SSI & SSO Data Plotting
do = squeeze(out.SSO.signals.values);
di = squeeze(out.SSI.signals.values);
t = out.SSO.time;

figure (1)
plot(t,do,'r-',t,di,'b-')
grid on
axis([0 0.000007 0 40])
xlabel('Time(sec)')
ylabel('Pressure (bar)')
title('Step Response')
legend('SSO', 'SSI')
```

7 Appendix D: Python Initial Sizing code

```
"""
The First Section of this code deals with the basic sizing of the engine and comes from
Richard Newlands book on hybrids
"""

#Constants
mu = 398600500 #gravitational potential
eradius=6378.14 #earths radius (kilometrea)
ullage=0.12
gzero=9.81 #sea level gravity

#Inputs
mass=10 #emptymass (kg)
apogee=3500 #maxaltitude (metres)
ce=2682 #exhaust velocity
ofrat=0.875 #o/f ratio

print("Calculation based on Newlands")
#Outputs
g=mu/((eradius+(apogee/1000))**2)
e=mu/(eradius+(apogee/1000)) #energy to apogee
v=math.sqrt((2*e)) #velocity required to reach apogee
testi=(apogee/v)/2 #estimated time of flight
gl=testi*g #gravity loss
tl=2*gl #total loss
deltav=v+tl #total deltaV needed
print("Delta V needed=", deltav)

mfrac=math.exp(deltav/ce) #mass fraction
print("Mass fraction=", mfrac)
propm1=(mfrac-1)*mass
print("Propellant Mass=", propm1)
oxim1=mfrac*(ofrat) #oxidiser mass
print("Oxidiser Mass=", oxim1)
fuelm1=propm1-oxim1 #fuelmass
print("Fuel Mass=", fuelm1)


""" This Second Section of code will use the more detailed methods
for sizing the rocket from Humble
"""
#Constants
finert=0.2 #inert mass fraction (Taken from Humble)
#Inputs
f=2000 #thrust (estimated)
isp=204.26 #from Cranfield 2024 report
print("\nCalculation based on Humble")
#Outputs
propm2 = (mass*(math.exp(deltav/(isp*gzero))-1)*(1-finert))/(1-finert*(math.exp(deltav/(isp*gzero))))
print("Propellant Mass=", propm2)
minert = (finert/(1-finert))*propm2
print("Inert Mass=", minert)
fuelm2 = propm2/(1+ofrat)
print("Fuel Mass=", fuelm2)
oxim2 = propm2-fuelm2
print("Oxidiser Mass=", oxim2)

wprop = f/(isp*gzero) #propellant mass flow
print("Propellant Massflow=", wprop)
wfuel = wprop/(1+ofrat) #fuel mass flow
print("Fuel Massflow=", wfuel)
woxi = wprop-wfuel #oxidiser mass flow
print("Oxidiser Massflow=", woxi)
```